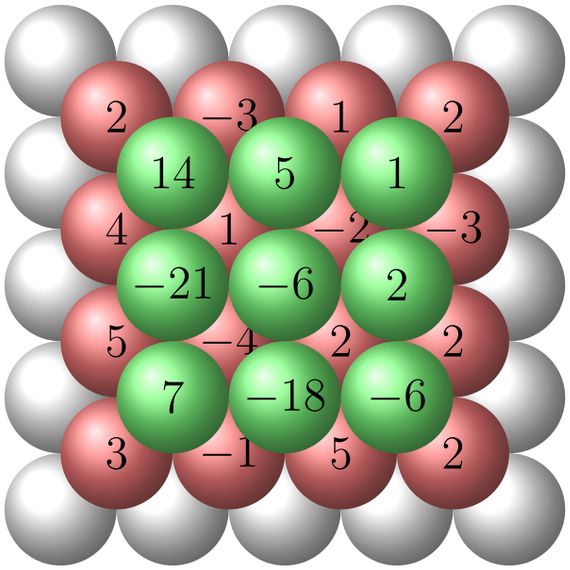
A rabbit hole of determinental identities

Greg Conradi Smith

[w/ Emily Simmons, Josh Puzey, Arielle Cooley]



If you are trying to say as much as you can about a square matrix using only one number, that number has got to be the *determinant*. —Gil Strang

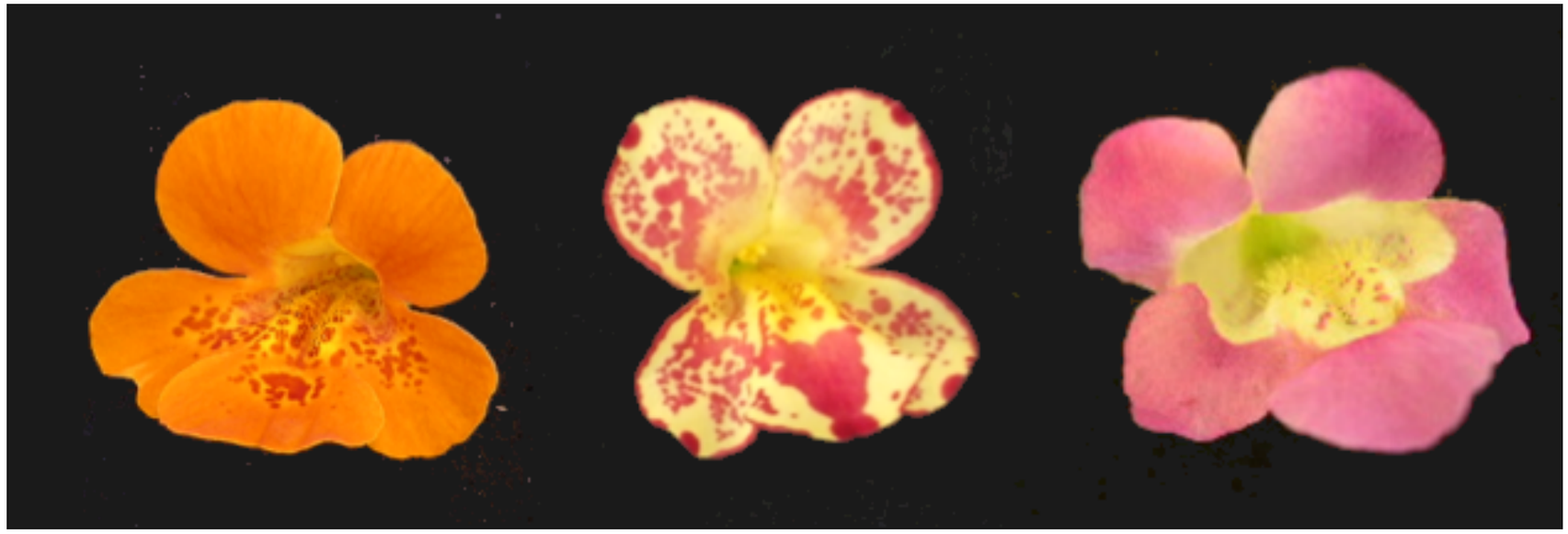
W&M biomath

2021 February 19



Hybridization creates novel phenotypes

M. cupreus F1 *M. variegatus*





Josh Puzey & Arielle Cooley

$$
\begin{aligned}
\frac{\partial u}{\partial t} &= D_u \nabla^2 u + f(u, v, \hat{u}, \hat{v}) \\
\frac{\partial v}{\partial t} &= D_v \nabla^2 v + g(u, v, \hat{u}, \hat{v}) \\
\frac{\partial \hat{u}}{\partial t} &= D_{\hat{u}} \nabla^2 \hat{u} + \hat{f}(u, v, \hat{u}, \hat{v}) \\
\frac{\partial \hat{v}}{\partial t} &= D_{\hat{v}} \nabla^2 \hat{v} + \hat{g}(u, v, \hat{u}, \hat{v})
\end{aligned}
$$

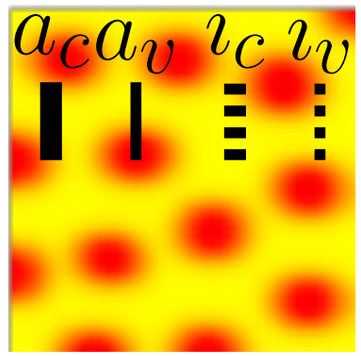
$$
A = \left[\begin{array}{cc|cc}
f_u & f_v & f_{\hat{u}} & f_{\hat{v}} \\
g_u & g_v & g_{\hat{u}} & g_{\hat{v}} \\
\hline
\hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\
\hat{g}_u & \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}}
\end{array}\right]
$$

Need to understand determinant of this 4-by-4 matrix in terms of 2-by-2 blocks

cupreus × *variegatus*

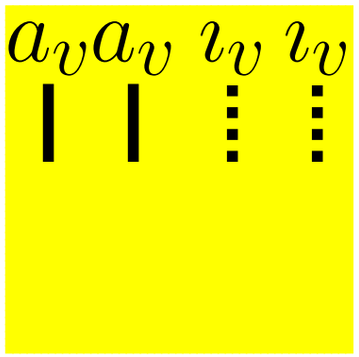


homozygous

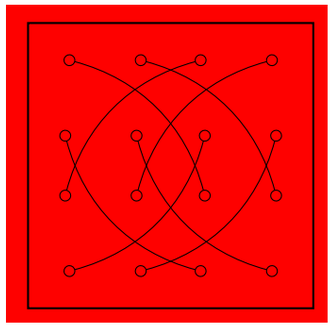


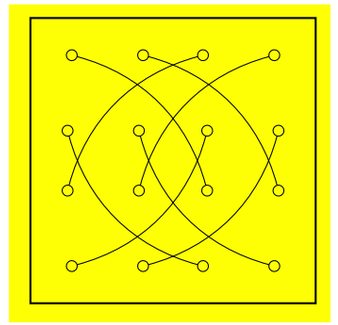
F1

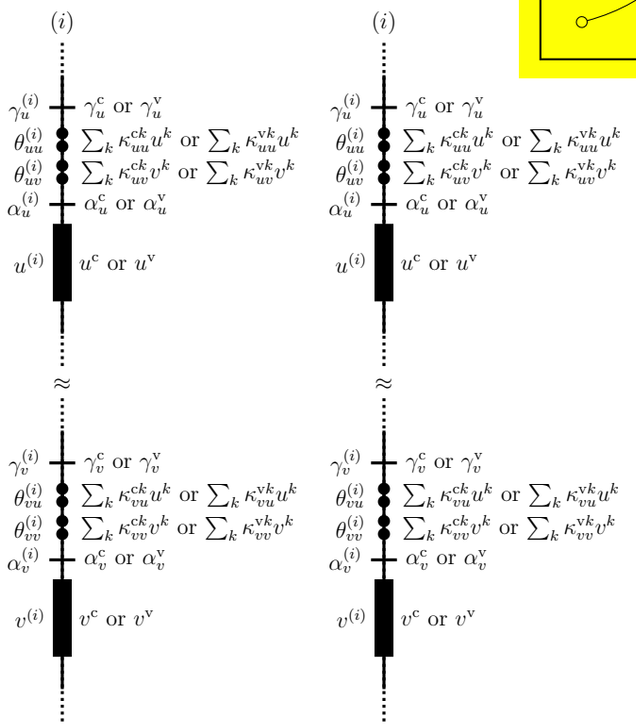
heterozygous



homozygous







In linear algebra, the determinant is a scalar value that can be computed from the elements of a square matrix.

The determinant encodes certain properties of the matrix and the linear transformation described by the matrix.

The determinant of a matrix A is denoted det(A), det A, or |A|.

In the case of a 2 × 2 matrix the determinant is

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

For a 3 × 3 matrix *A*, its determinant is

$$|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - bdi - afh.$$

$$\det A = |A| = 0 \iff A \text{ is singular}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \iff \begin{bmatrix} a \\ c \end{bmatrix} = \alpha \begin{bmatrix} b \\ d \end{bmatrix} \iff \begin{array}{c} a = \alpha b \\ \& \\ c = \alpha d \end{array}$$

singular

$$\Updownarrow$$

$$ad - bc = 0 \iff \alpha = \frac{c}{d} \;\&\; a = \frac{c}{d}\, b$$

$$|A| = ad - bc$$

$$\det A = |A| = 0 \iff A \text{ is singular}$$

$$\underbrace{\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}}_{\text{singular}} \iff \underbrace{\begin{bmatrix} a & b & c \\ 0 & e - \frac{bd}{a} & f - \frac{cd}{a} \\ 0 & h - \frac{bg}{a} & i - \frac{cg}{a} \end{bmatrix}}_{\text{singular}}$$

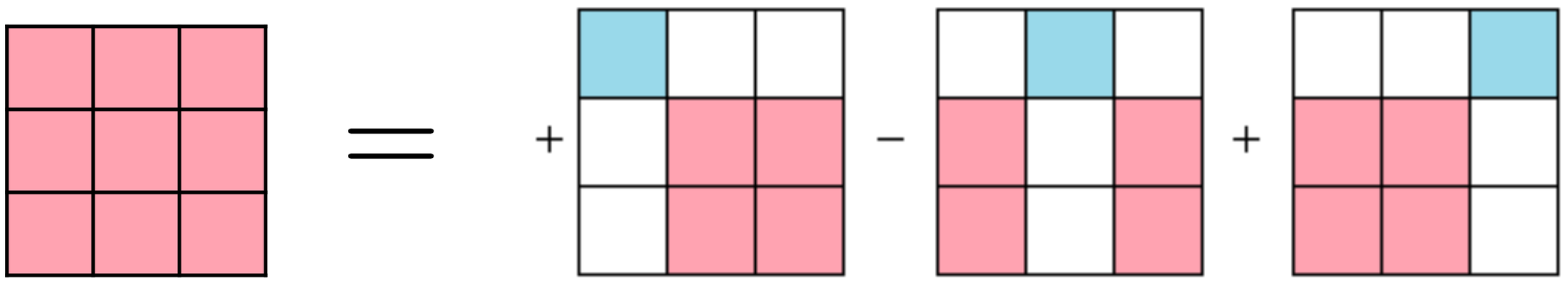
$$\Updownarrow$$

$$\begin{vmatrix} e - \frac{bd}{a} & f - \frac{cd}{a} \\ h - \frac{bg}{a} & i - \frac{cg}{a} \end{vmatrix} = 0$$

$$\left(e - \frac{bd}{a}\right)\left(i - \frac{cg}{a}\right) - \left(f - \frac{cd}{a}\right)\left(h - \frac{bg}{a}\right) = 0$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei - afh + bfg - bdi + cdh - ceg = 0$$

Laplace expansion of a determinant



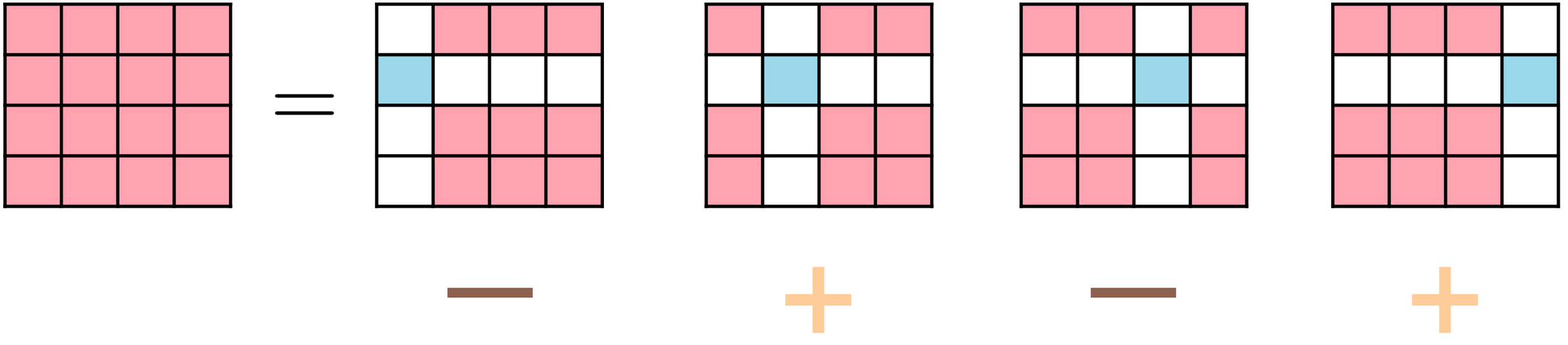
$$|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} \square & \square & \square \\ \square & e & f \\ \square & h & i \end{vmatrix} - b \begin{vmatrix} \square & \square & \square \\ d & \square & f \\ g & \square & i \end{vmatrix} + c \begin{vmatrix} \square & \square & \square \\ d & e & \square \\ g & h & \square \end{vmatrix}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

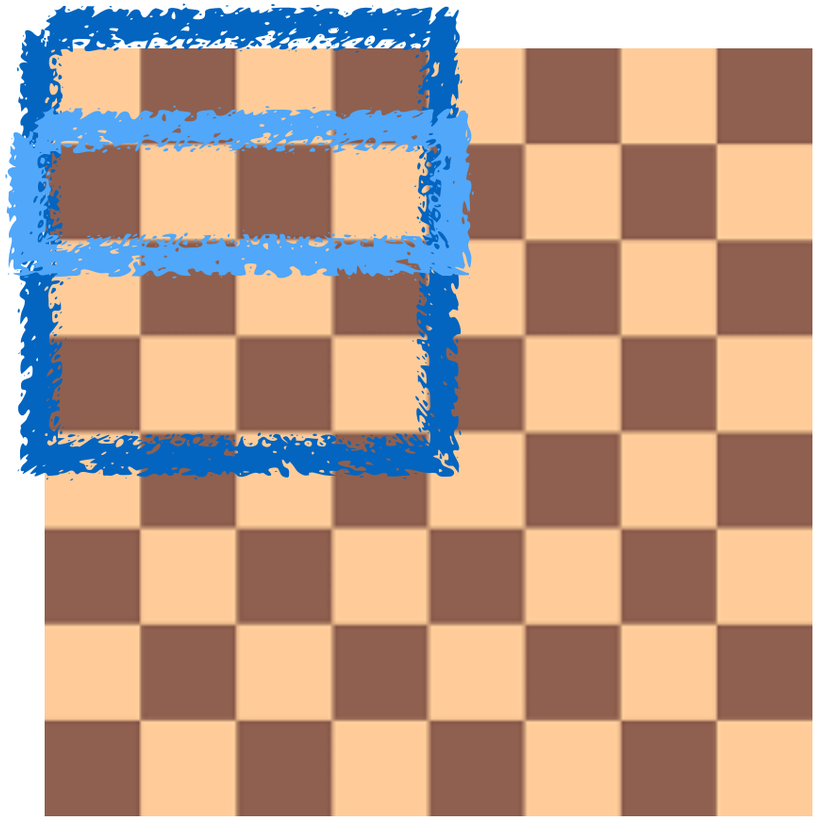
$$= aei + bfg + cdh - ceg - bdi - afh.$$

David Eberly

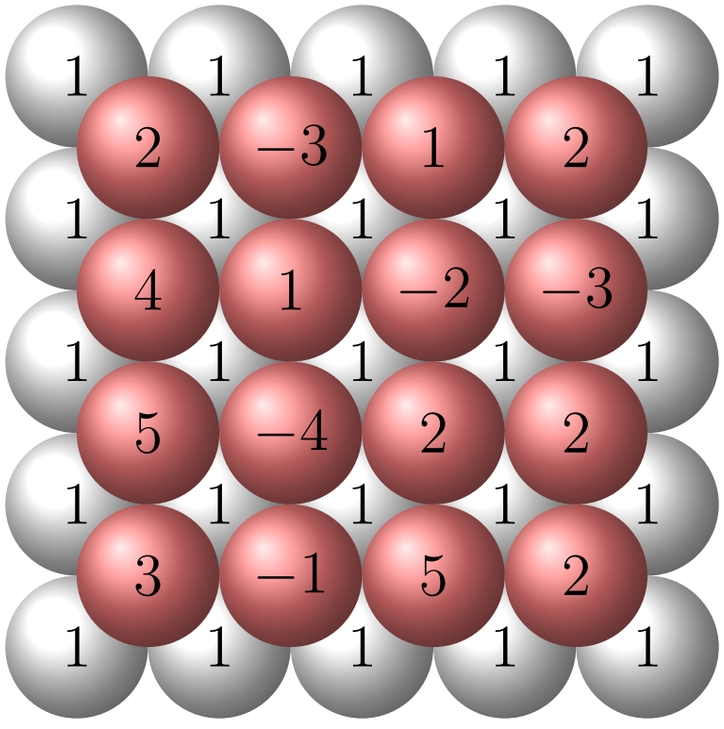
Laplace expansion of a determinant

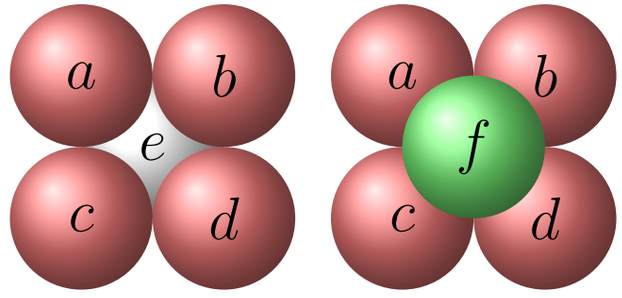


$$\begin{vmatrix} 2 & 3 & 5 & 1 \\ 0 & 0 & 7 & 0 \\ 1 & -4 & 13 & 1 \\ 0 & 11 & 17 & 0 \end{vmatrix} = 77$$

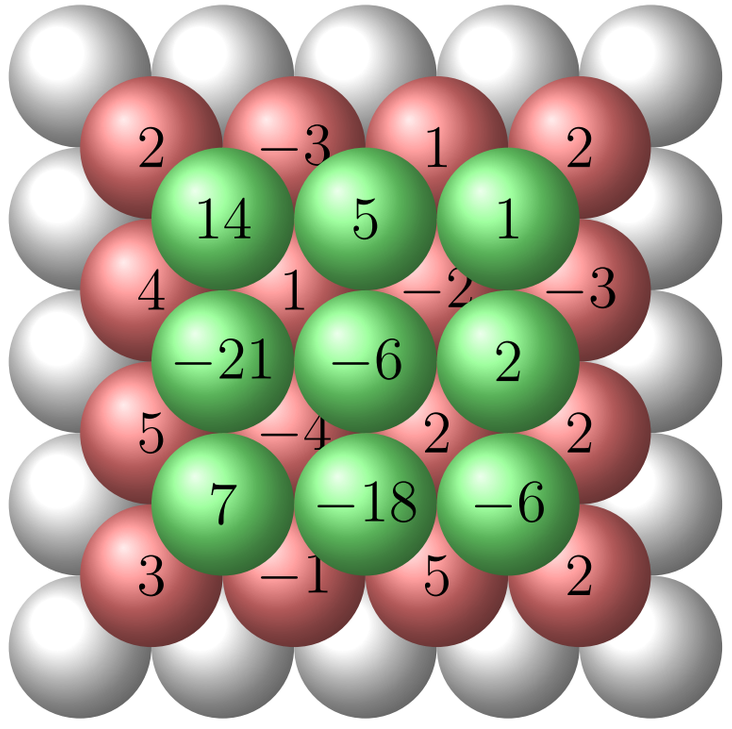


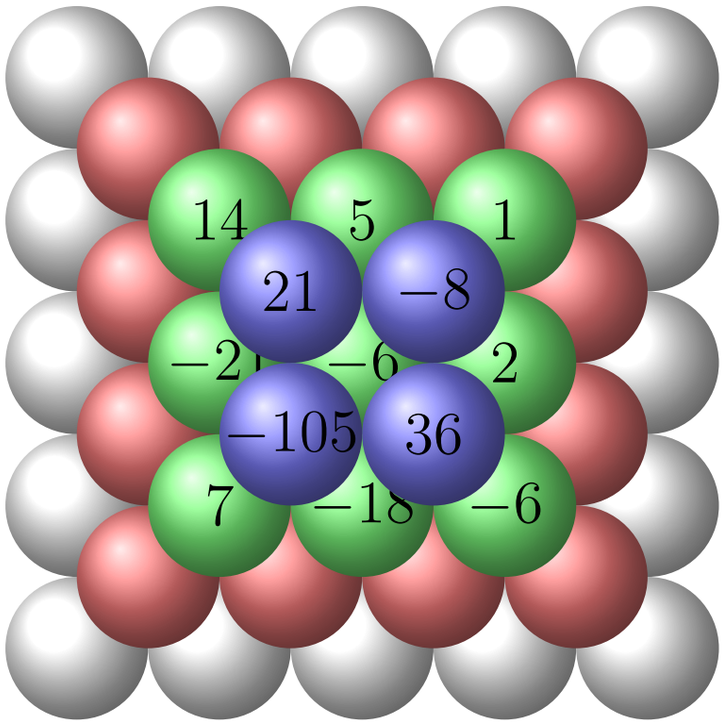
Dodgson condensation

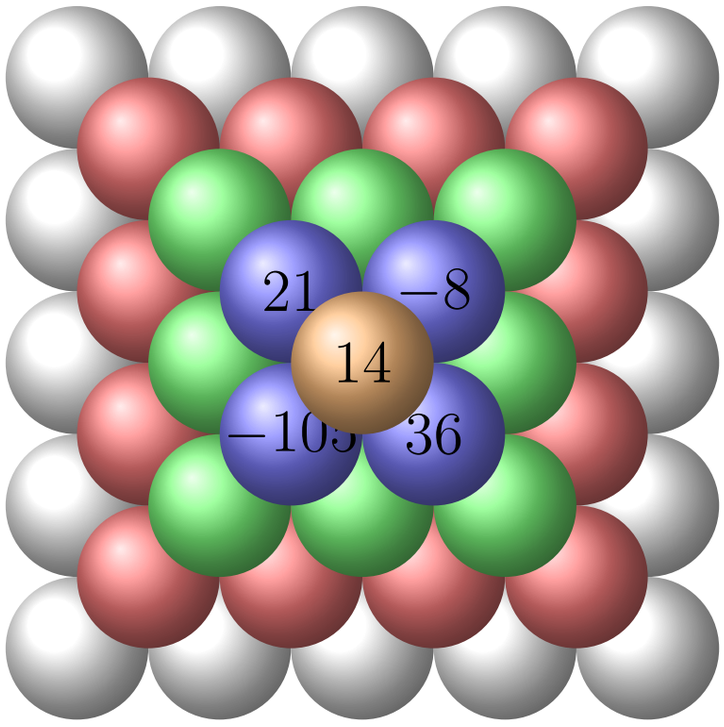




$$f = \frac{ad - bc}{e}$$







I made this using TikZ

The determinant encodes certain properties of the matrix and the linear transformation described by the matrix

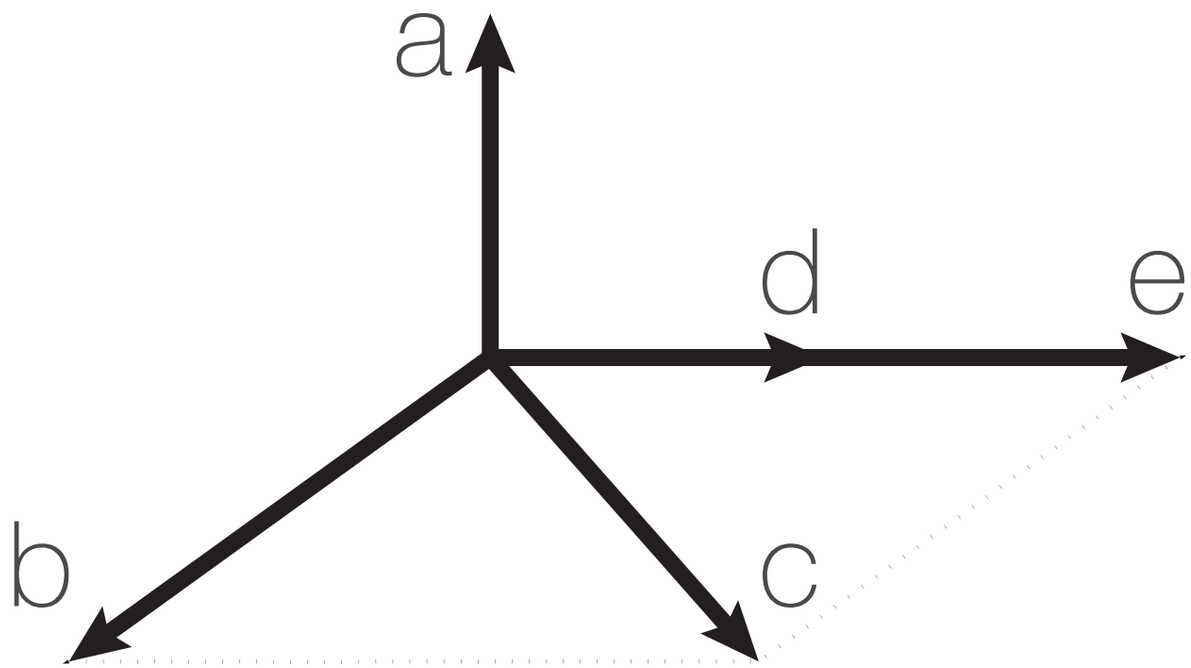




- |A| = 0 iff A is singular (not invertible).
- |A| ≠ 0 iff the columns of A are linearly independent
- |A| = signed area associated to columns of A
- |A| = $\lambda_1 \lambda_2 \cdots \lambda_n$ (the product of the eigenvalues of A)
- In two-dimensional linear systems,

$$\frac{d\boldsymbol{x}}{dt} = A\boldsymbol{x} \Longleftrightarrow \mathrm{tr}A < 0 \ \& \ \det A > 0$$

independent vs. dependent vectors



independent sets of vectors

{}

{a},{b},{c},{d},{e}

{a,b},{a,c},{a,d},{a,e},{b,c},
{b,d},{b,e},{c,d},{c,e}

{a,b,c}, {a,b,d}, {a,b,e},
{a,b,c}, {a,b,d}, {a,b,e},

dependent sets of vectors

{d,e}

{a,d,e},{b,c,d},{b,c,e},
{b,d,e},{c,d,e}

{a,b,c,d},{a,b,c,e},{a,b,d,e},
{a,c,d,e},{b,c,d,e}

{a,b,c,d,e}

Are $\begin{bmatrix}2\\2\\1\end{bmatrix}, \begin{bmatrix}-4\\6\\5\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}$ linearly independent?

→ Independent if $\alpha_1\begin{bmatrix}2\\2\\1\end{bmatrix} + \alpha_2\begin{bmatrix}-4\\6\\5\end{bmatrix} + \alpha_3\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$

has only the trivial solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$

Adam Panagos

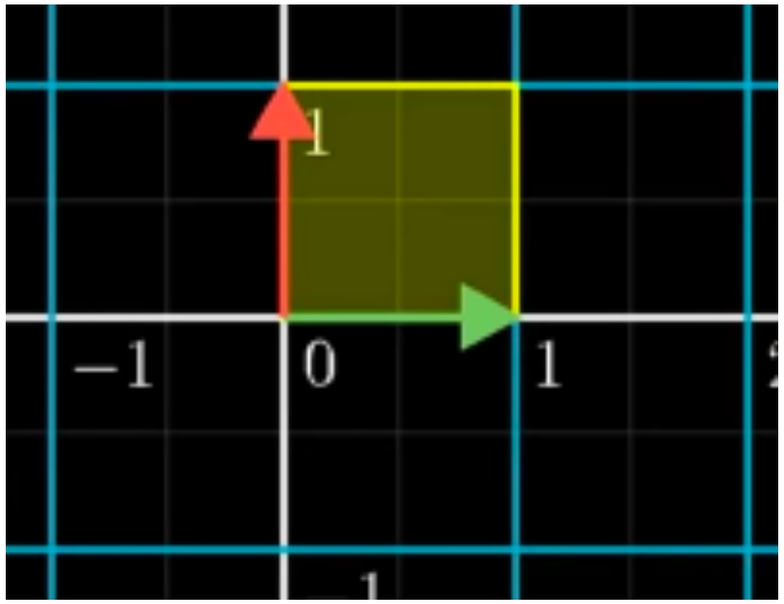
Otherwise the vectors are dependent

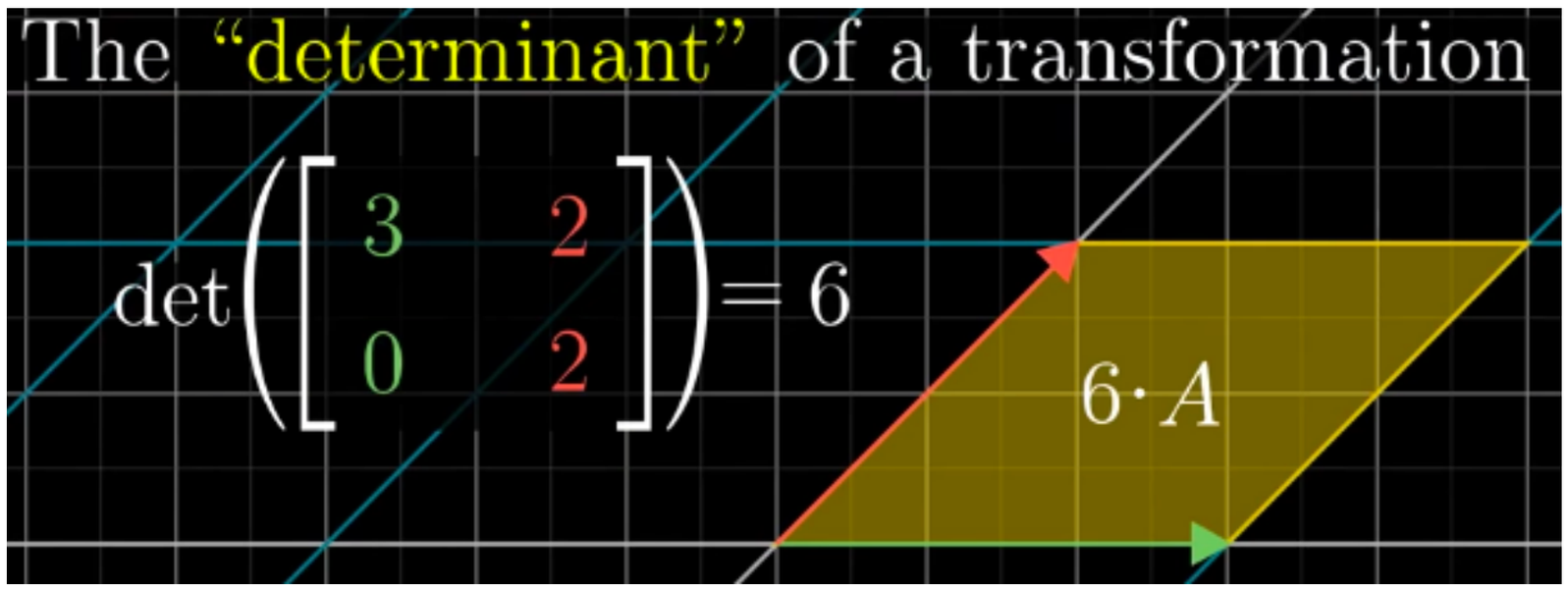
$$\begin{bmatrix}2 & -4 & 1 & 0\\ 2 & 6 & 0 & 0\\ 1 & 5 & 0 & 0\end{bmatrix} \xrightarrow{r.r.} \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix} \Rightarrow \begin{aligned}\alpha_1 &= 0\\ \alpha_2 &= 0\\ \alpha_3 &= 0\end{aligned}$$

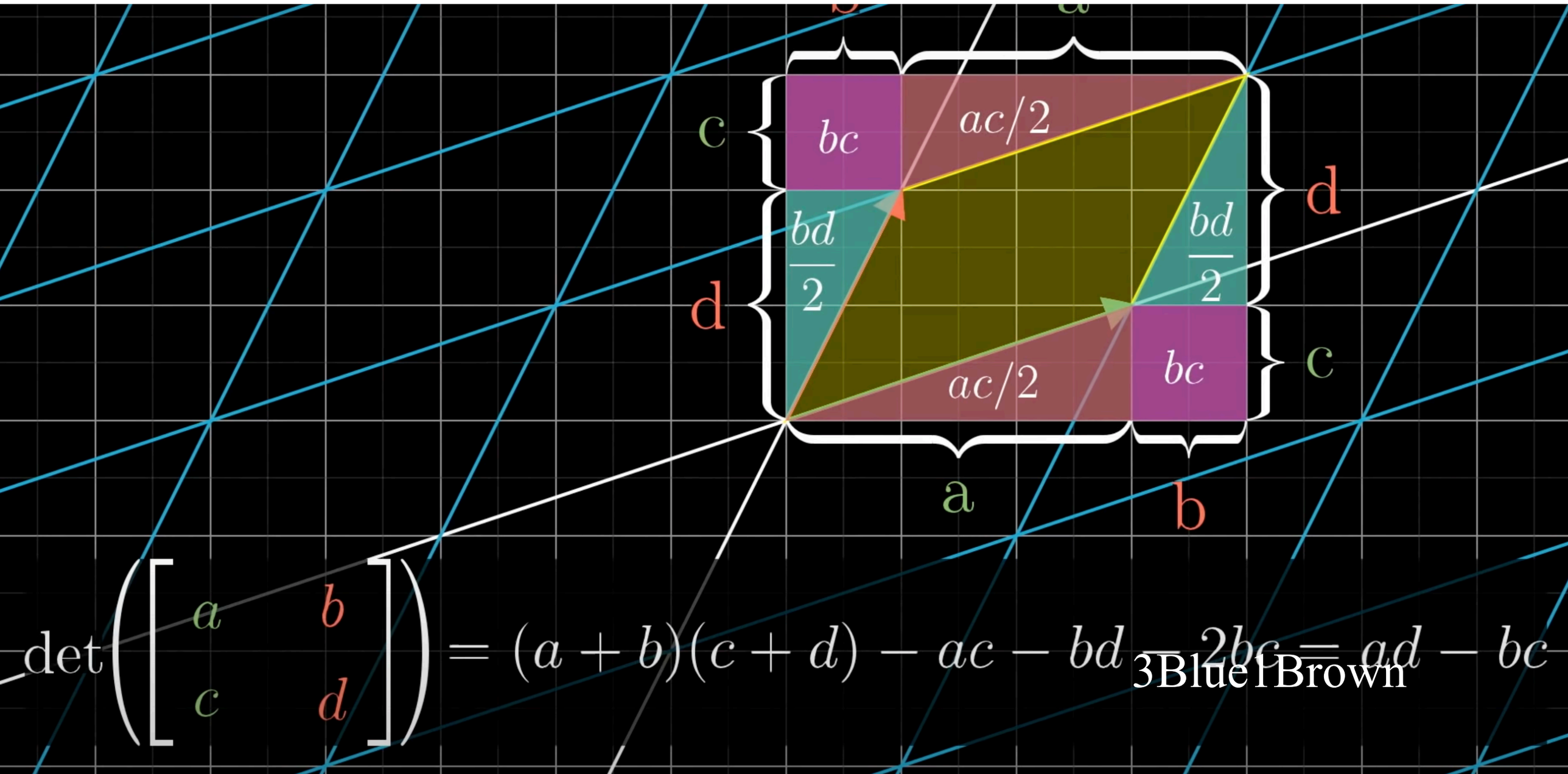
Augmented matrix

```
>> det([2 2 1; -4 6 5; 1 0 0 ])

ans = 4.0000
```


1
−1
0
1

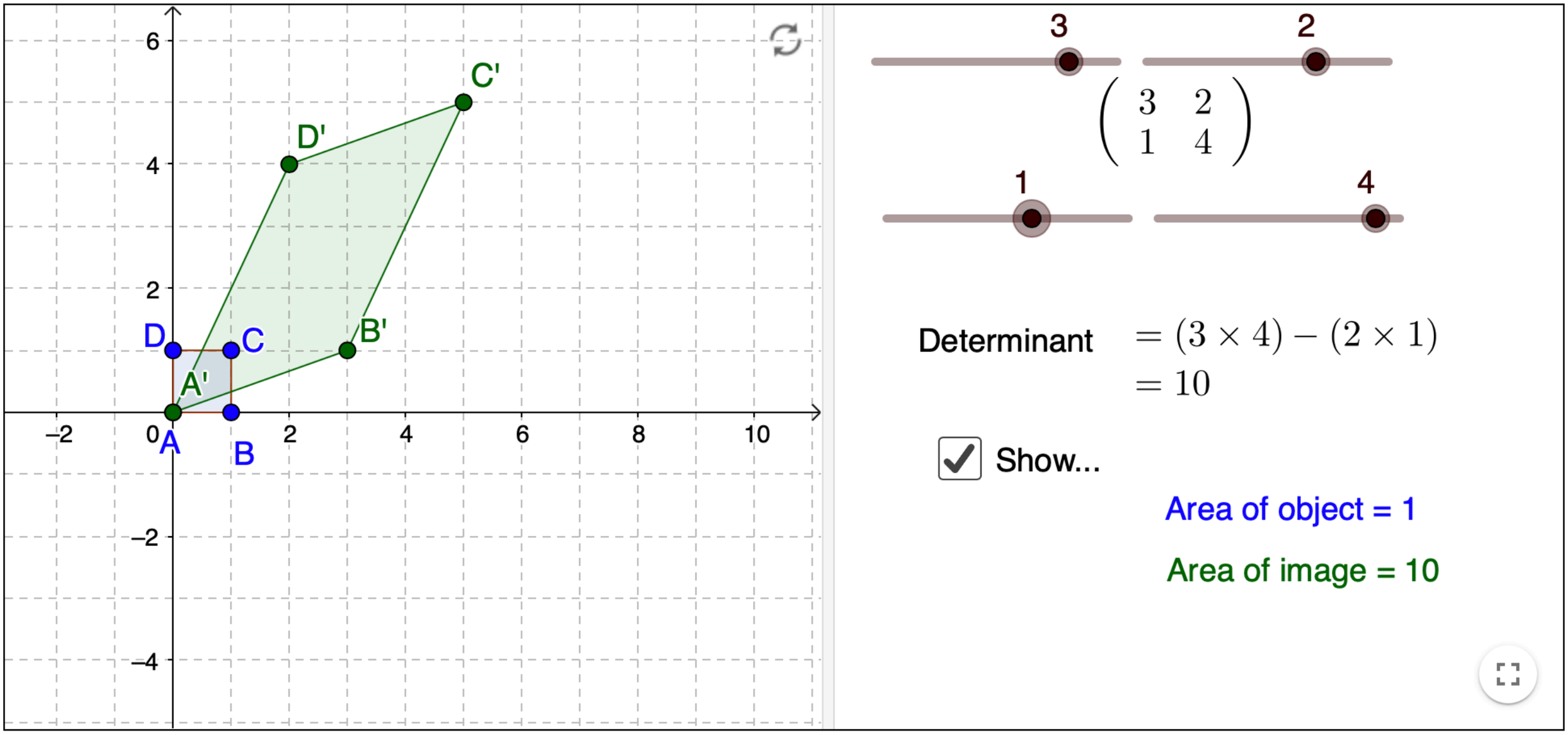

The "determinant" of a transformation
$\det\left(\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}\right) = 6$
$6 \cdot A$


b
a
c
bc
$ac/2$
d
$\frac{bd}{2}$
d
$\frac{bd}{2}$
c
$ac/2$
bc
a
b
$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b)(c+d) - ac - bd - 2bc = ad - bc$
3Blue1Brown

Determinants and area for linear transformations

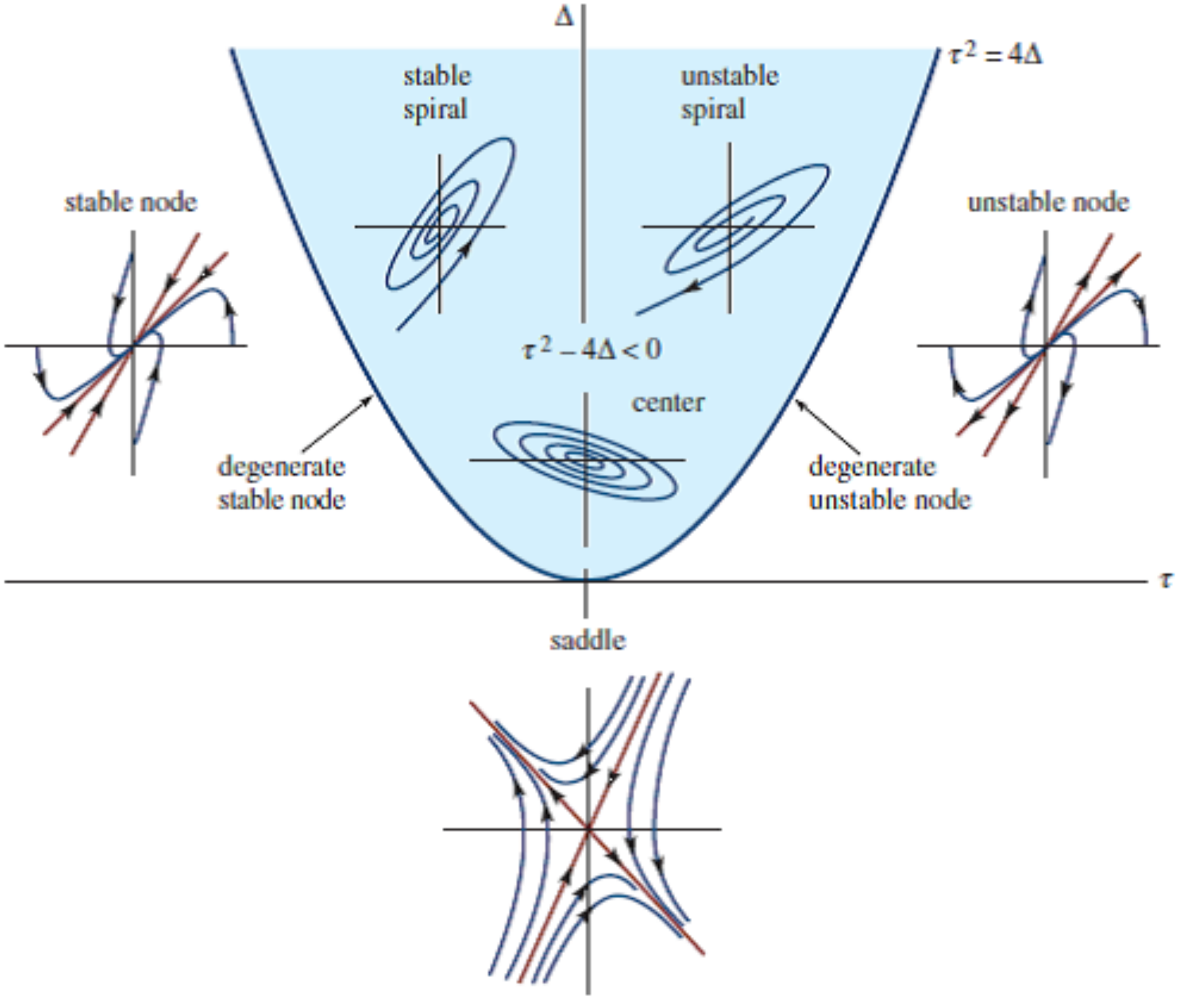
Author: Antonella Zanna, Catherine Berry

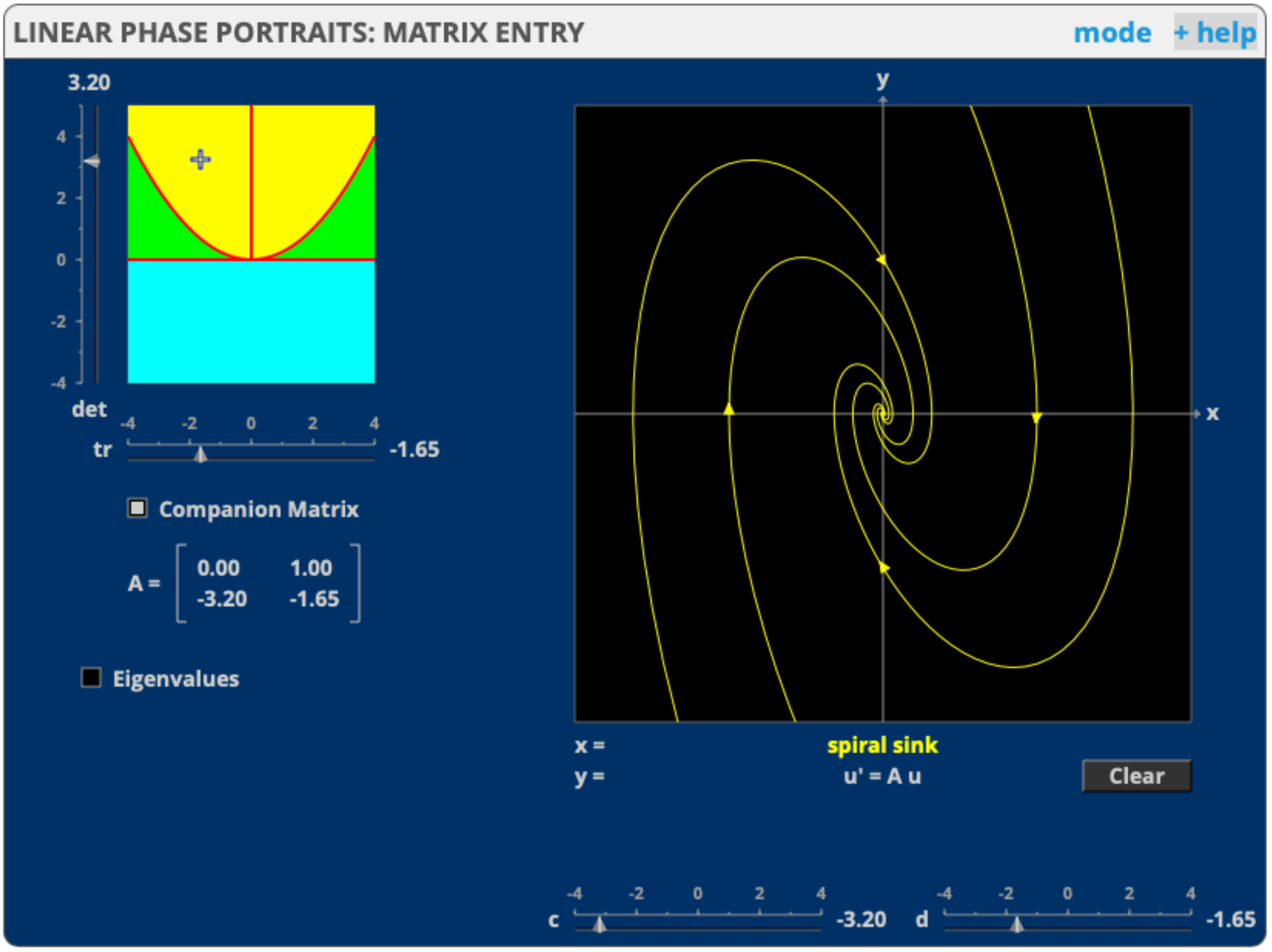
Topic: Area



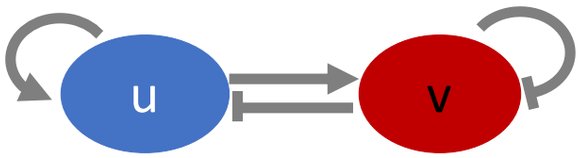
https://www.geogebra.org/m/H2xbpuEa

trace-determinant plane



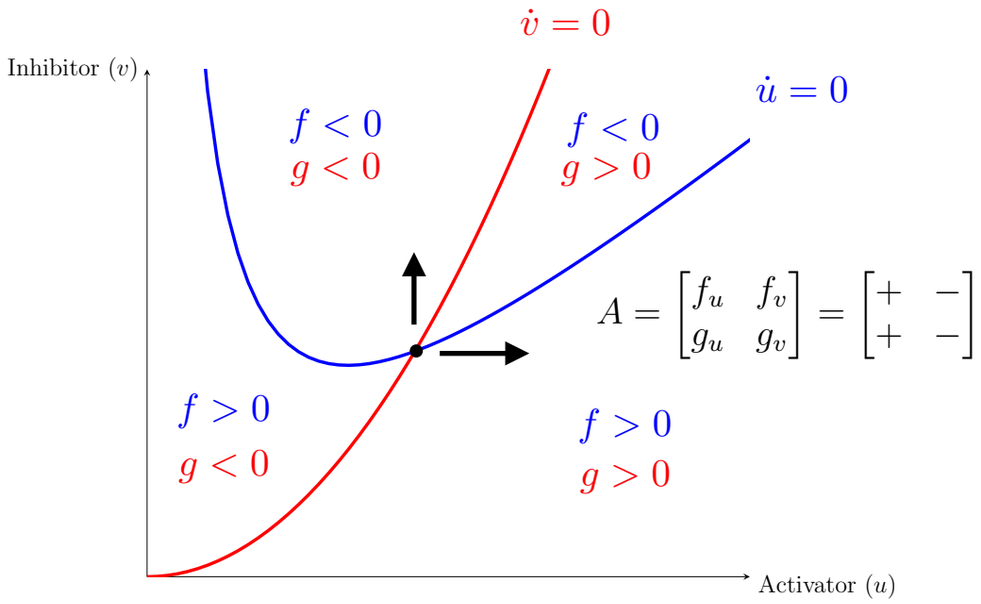


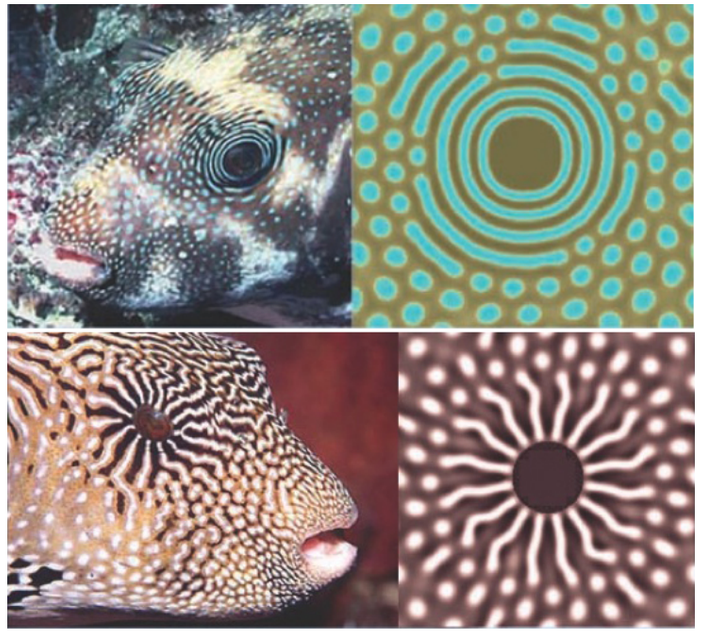
https://mathlets.org/mathlets/linear-phase-portraits-matrix-entry/

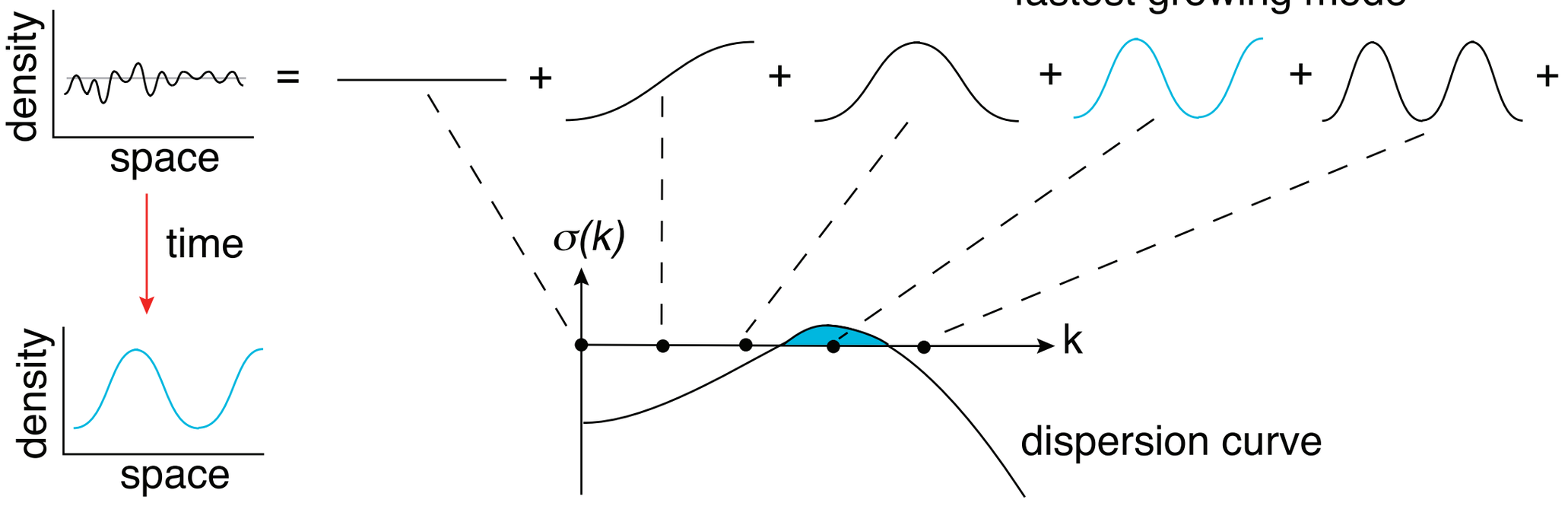


$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u,v)$$

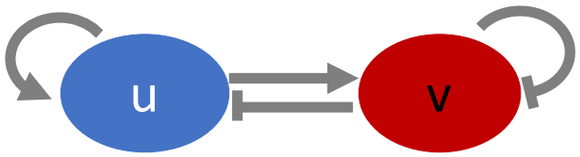
$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u,v)$$







Spatially homogenous steady state is stable without diffusion.
But system is unstable to spatial perturbations with Dv > Du



$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u,v)$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u,v)$$

$$\frac{\partial}{\partial t}\begin{pmatrix} U \\ V \end{pmatrix} = \mathbb{L}\begin{pmatrix} U \\ V \end{pmatrix} \equiv \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}_{u^*,v^*}\begin{pmatrix} U \\ V \end{pmatrix} + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}\begin{pmatrix} \nabla^2 U \\ \nabla^2 V \end{pmatrix}, \tag{11.2.2}$$

supplemented by no-flux boundary conditions. Here $f_u = \partial f/\partial u$ etc. Setting $\mathbf{U} = (U,V)^T$, the general solution of (11.2.2) can be written as

$$\mathbf{U}(\mathbf{x},t) = \sum_{\mathbf{k}} \mathbf{c}_k \mathrm{e}^{i\mathbf{k}\cdot\mathbf{x}} \mathrm{e}^{\lambda(k)t},$$

where $\mathbf{c}_k \mathrm{e}^{i\mathbf{k}\cdot\mathbf{x}}$ and $\lambda(k)$ form an eigenvalue pair of the linear operator $\mathbb{L}$, parameterized by the wavenumber $k = |\mathbf{k}|$. The k-dependence of the eigenvalue $\lambda(k)$ is known as a dispersion relation. Substitution of the general solution into (11.2.2) yields a characteristic equation for $\lambda(k)$:

$$\left|\mathbf{A} - \mathbf{D}k^2 - \lambda(k)\mathbf{I}\right| = 0, \tag{11.2.3}$$

with

$$\mathbf{A} = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}_{u^*,v^*}, \quad \mathbf{D} = \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}.$$

Evaluating the determinant, we obtain a quadratic equation for λ:

$$\lambda^2 + [(D_u + D_v)k^2 - f_u - g_v]\lambda + D_u D_v k^4 - k^2(D_v f_u + D_u g_v) + f_u g_v - f_v g_u = 0.$$

In the absence of diffusion ($D_u = D_v = 0$), this reduces to

$$\lambda^2 - [f_u + g_v]\lambda + f_u g_v - f_v g_u = 0,$$

and the requirement that the homogeneous state is stable in the absence of diffusion leads to the conditions

$$f_u + g_v < 0, \quad f_u g_v - f_v g_u > 0. \tag{11.2.4}$$

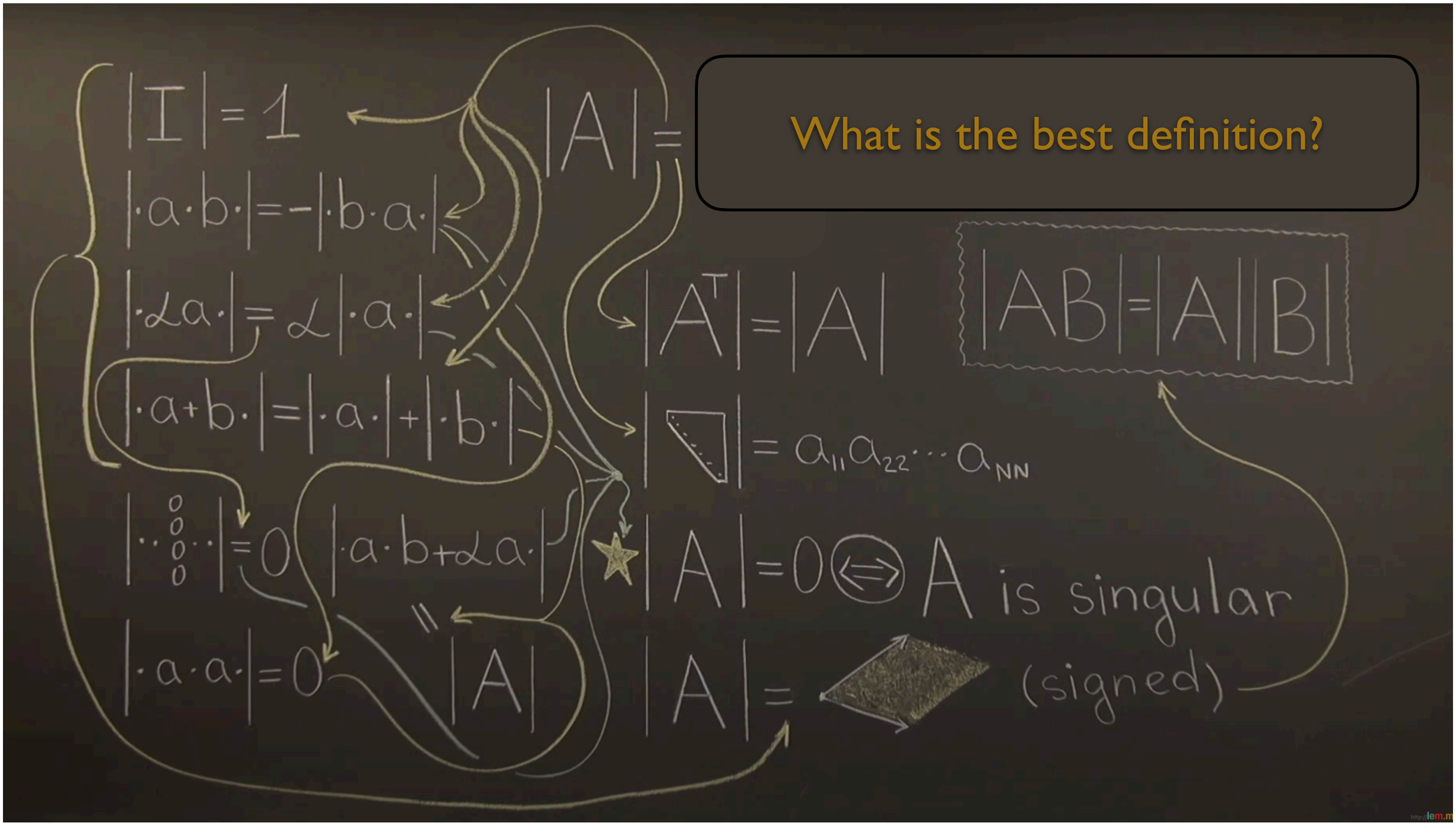
On the other hand, the requirement that the stationary state is unstable to perturbations in the presence of diffusion means that there exists a non-zero wavenumber k for which $\lambda(k) = 0$. Setting $\lambda = 0$ in the quadratic equation yields

$$D_u D_v k^4 - k^2(D_v f_u + D_u g_v) + f_u g_v - f_v g_u = 0,$$

which will have a positive solution for k^2 provided that

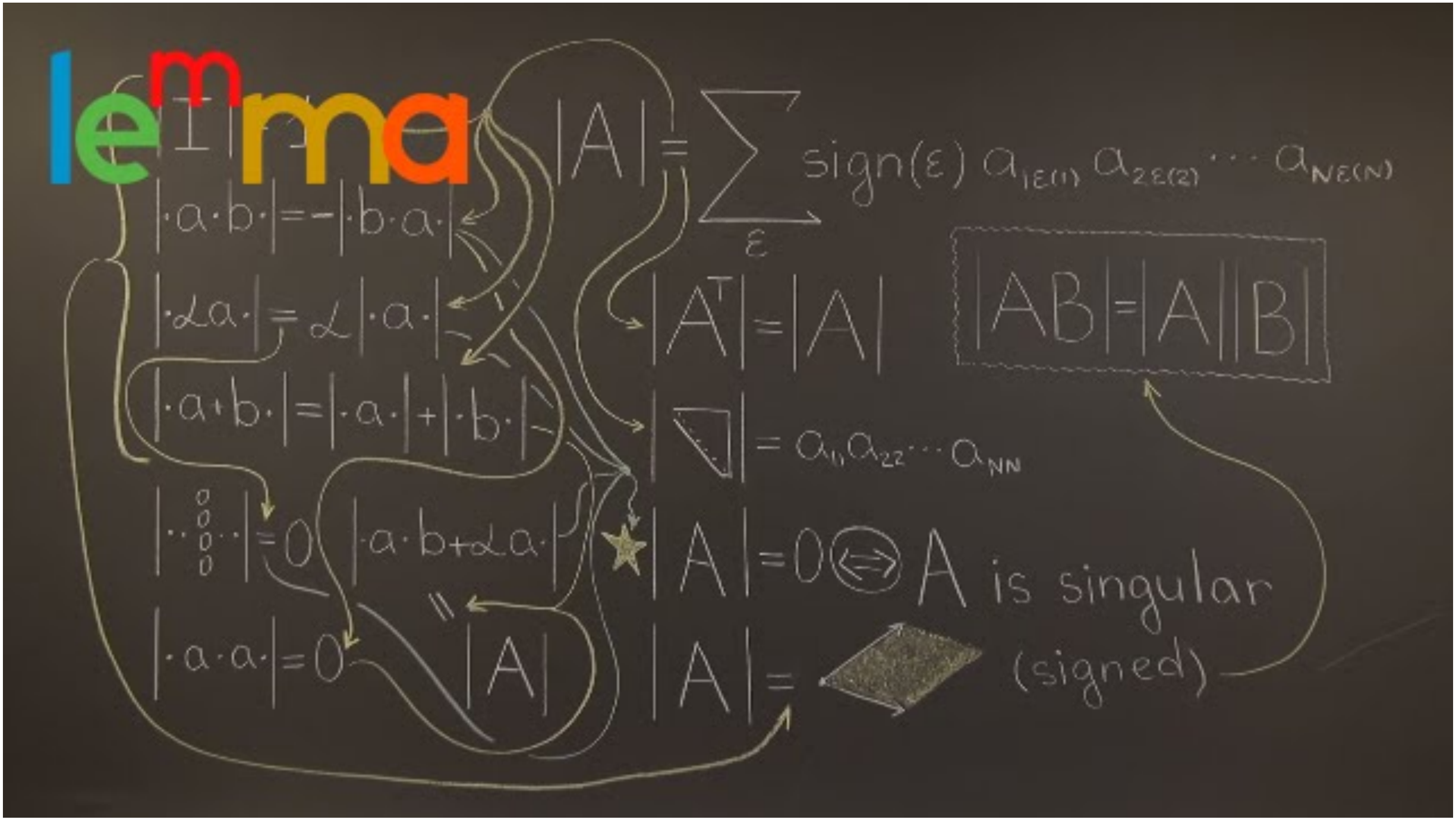
$$D_v f_u + D_u g_v > 0. \tag{11.2.5}$$

properties vs. definition of the determinant



Pavel Grinfeld

If you are trying to say as much as you can about a square matrix using only one number, that number has got to be the *determinant*. —Gil Strang



Pavel Grinfeld

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\det(A) = \sum_{\sigma \in S_n} \left(\operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma_i} \right)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

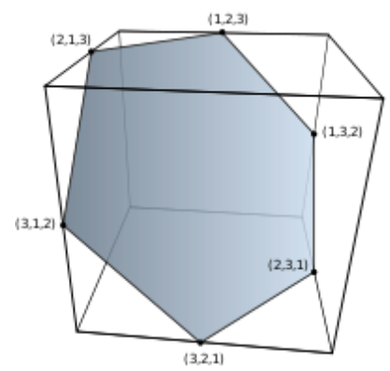
$$+a_{11}a_{22}a_{33}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$-a_{12}a_{21}a_{33}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$-a_{11}a_{23}a_{32}$$



$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$+a_{12}a_{23}a_{31}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$-a_{13}a_{22}a_{31}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

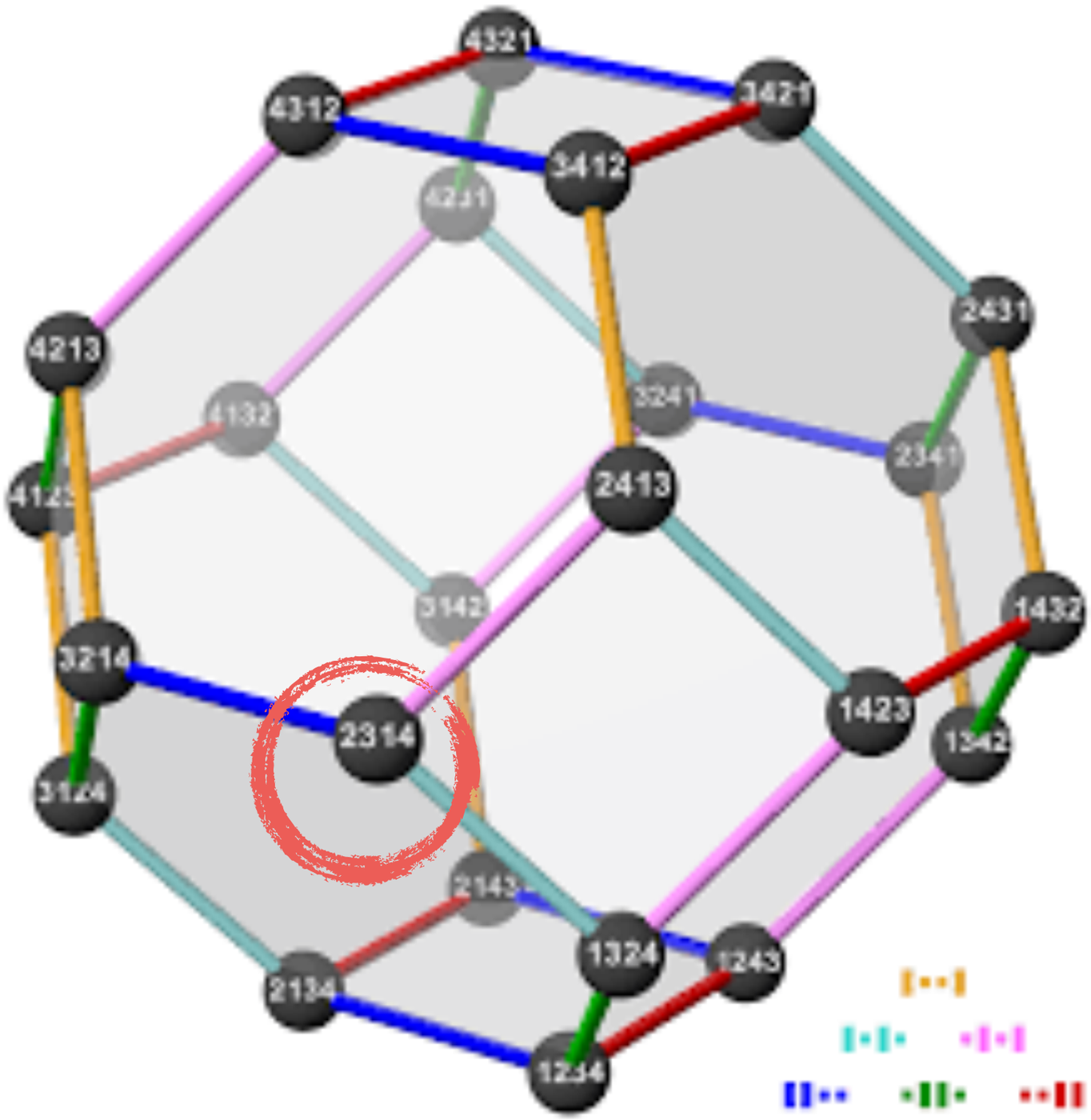
$$+a_{13}a_{21}a_{32}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

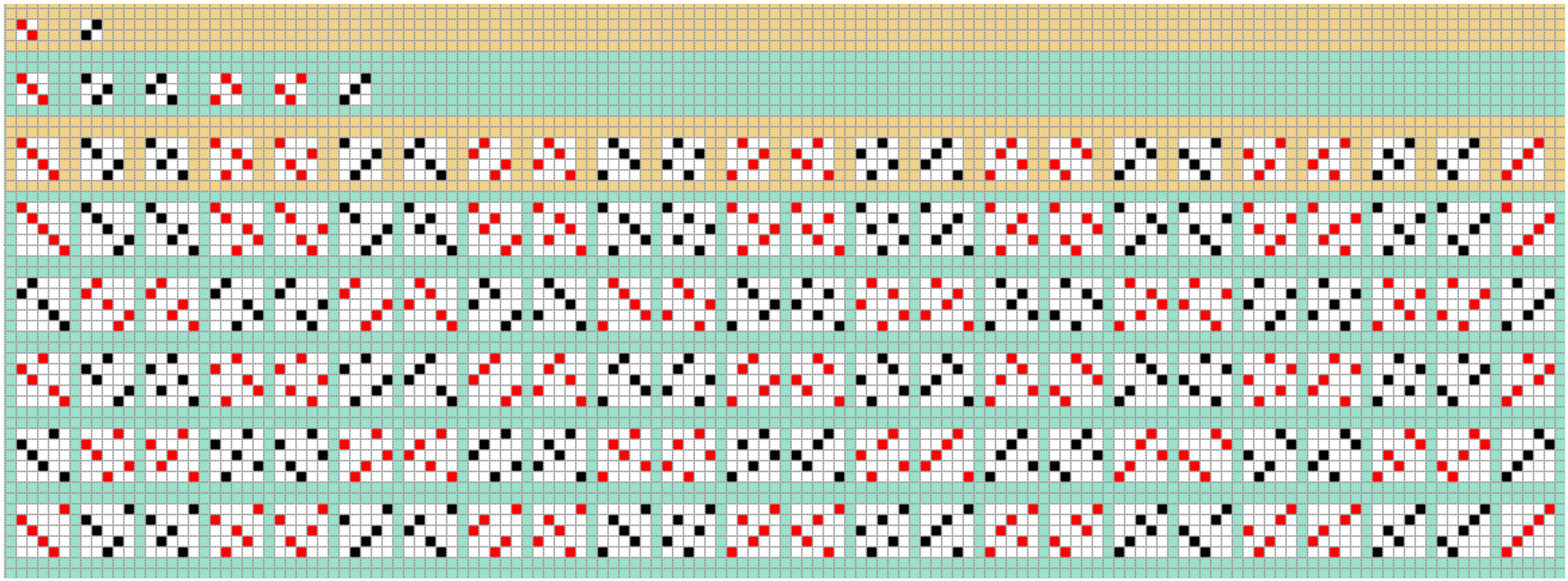
$$+a_{12}a_{23}a_{31}a_{34}$$

The permutohedron of order 4
(a 3-dimensional polytope)



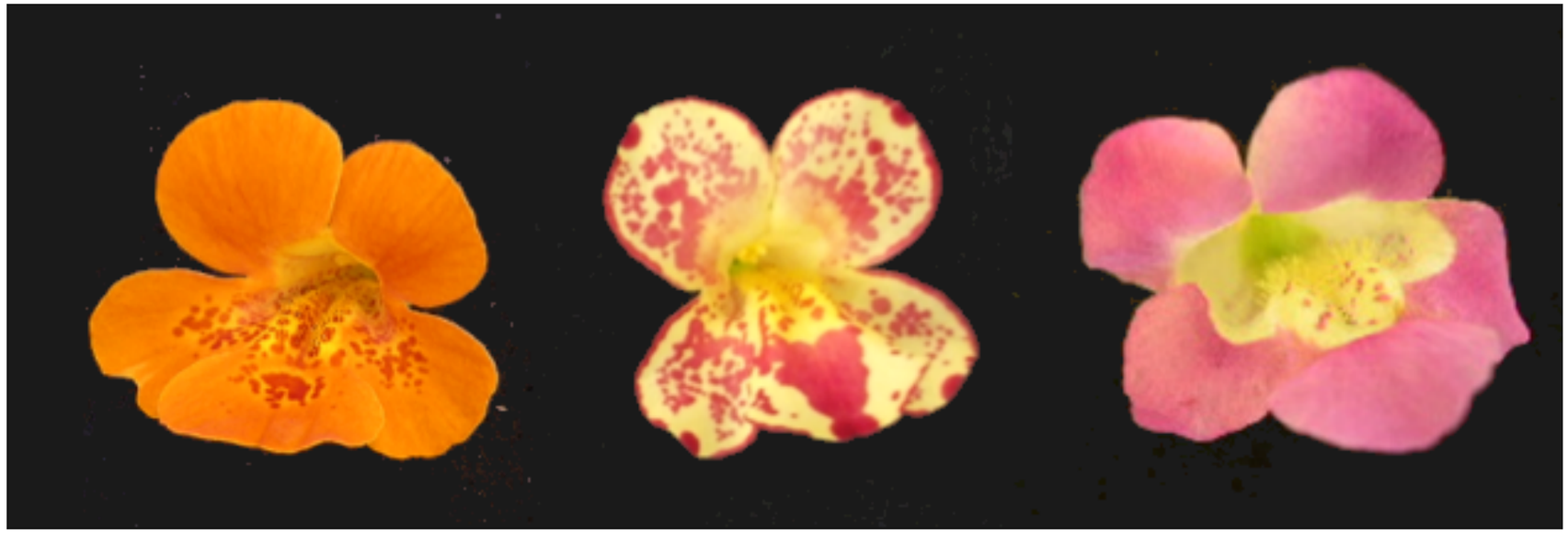


The determinant of an n-by-n matrix is a homogenous polynomial of degree n with n! terms half of which are negative



Hybridization creates novel phenotypes

M. cupreus F1 *M. variegatus*





Josh Puzey & Arielle Cooley

$$
\begin{aligned}
\frac{\partial u}{\partial t} &= D_u \nabla^2 u + f(u, v, \hat{u}, \hat{v}) \\
\frac{\partial v}{\partial t} &= D_v \nabla^2 v + g(u, v, \hat{u}, \hat{v}) \\
\frac{\partial \hat{u}}{\partial t} &= D_{\hat{u}} \nabla^2 \hat{u} + \hat{f}(u, v, \hat{u}, \hat{v}) \\
\frac{\partial \hat{v}}{\partial t} &= D_{\hat{v}} \nabla^2 \hat{v} + \hat{g}(u, v, \hat{u}, \hat{v})
\end{aligned}
$$

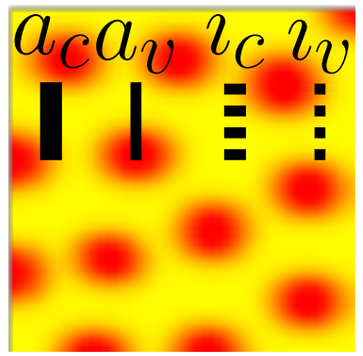
$$
A = \left[\begin{array}{cc|cc}
f_u & f_v & f_{\hat{u}} & f_{\hat{v}} \\
g_u & g_v & g_{\hat{u}} & g_{\hat{v}} \\
\hline
\hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\
\hat{g}_u & \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}}
\end{array}\right]
$$

Want to determine stability of this 4-by-4 matrix using information about stability of 2-by-2 blocks

cupreus × *variegatus*

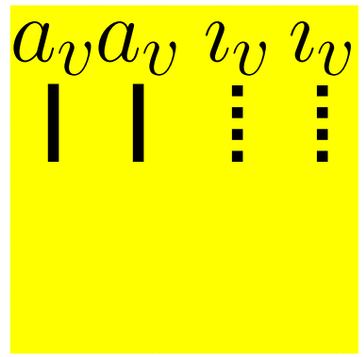


homozygous

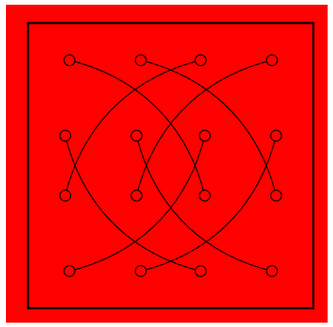


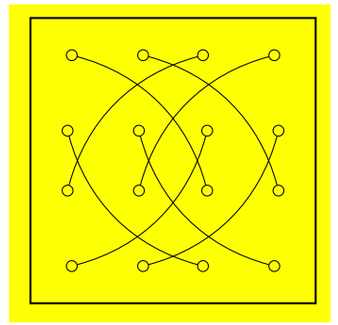
F1

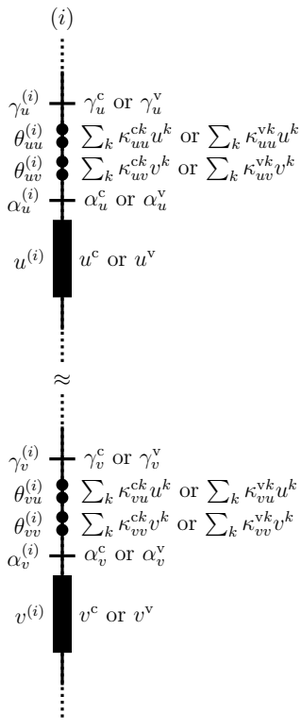
heterozygous

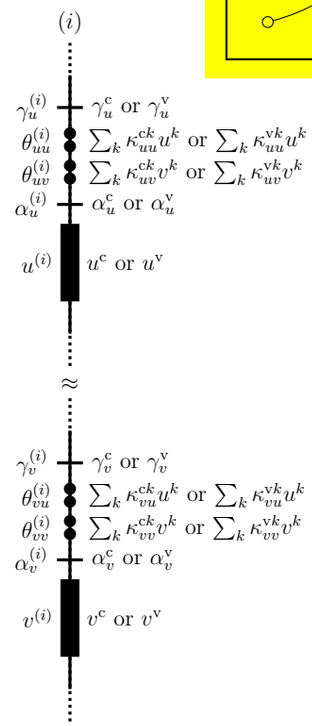


homozygous

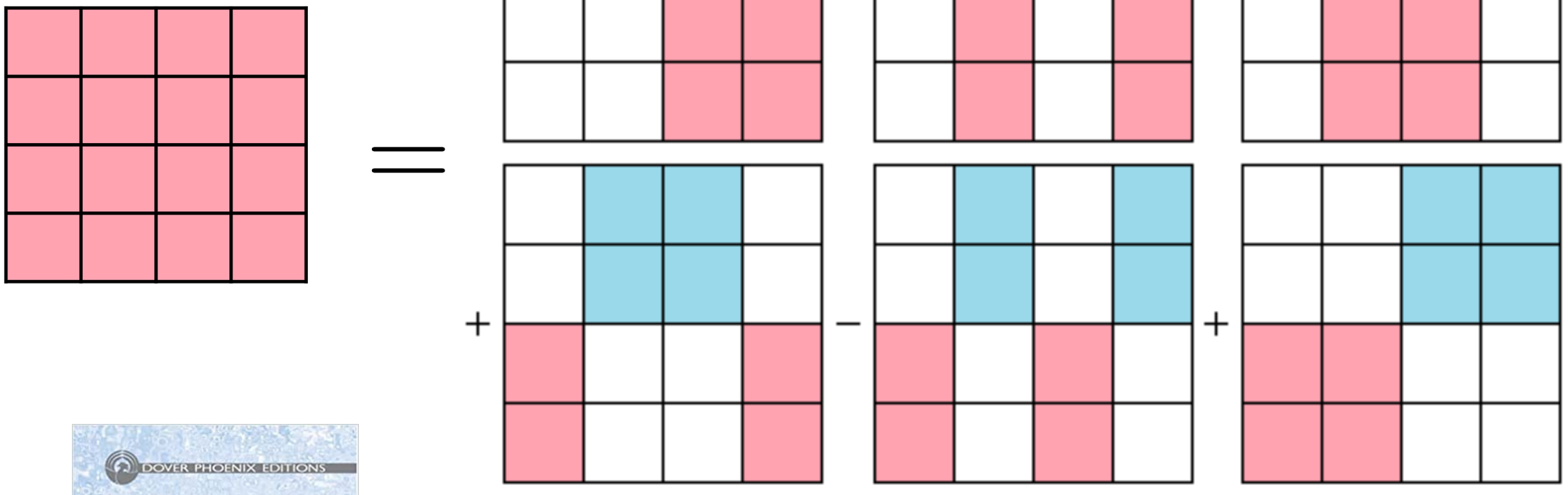


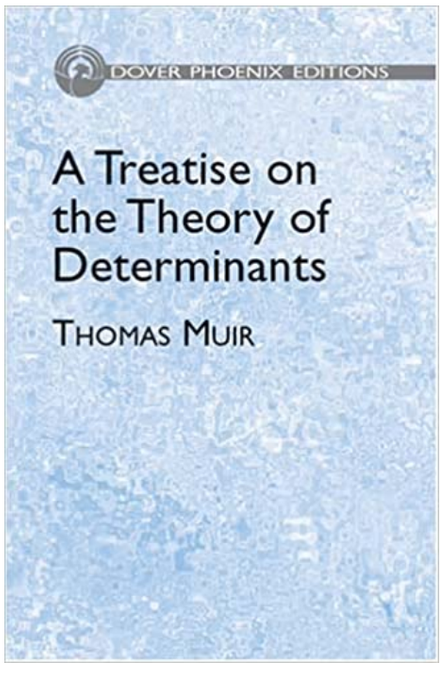






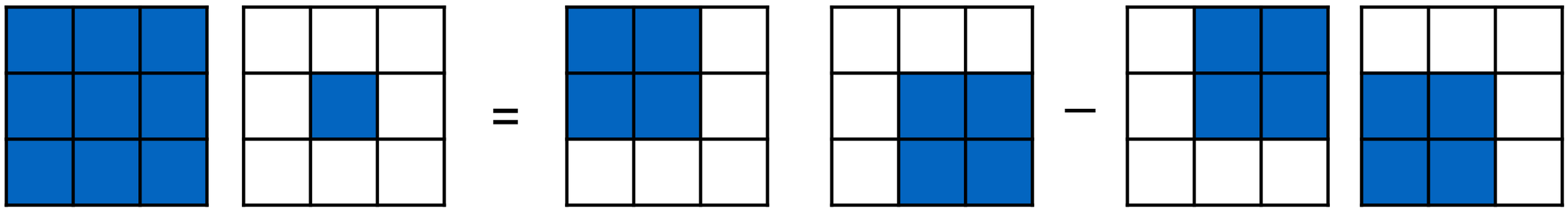
Laplace expansion of a determinant that respects block structure





Dodgson's Rule

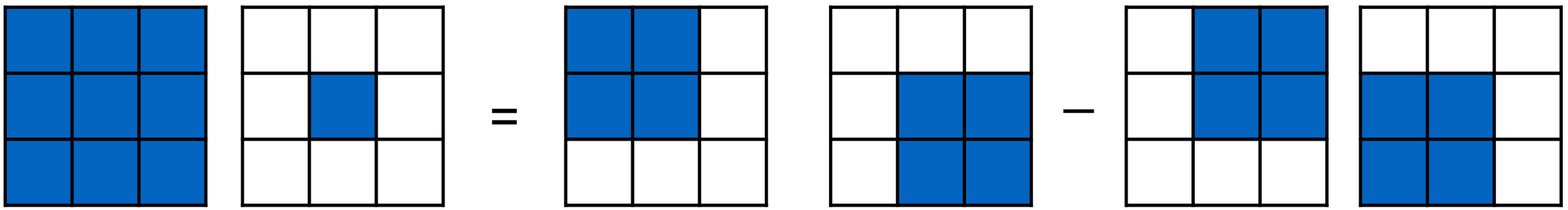
$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} e = \begin{vmatrix} a & b \\ d & e \end{vmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} - \begin{vmatrix} b & c \\ e & f \end{vmatrix} \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

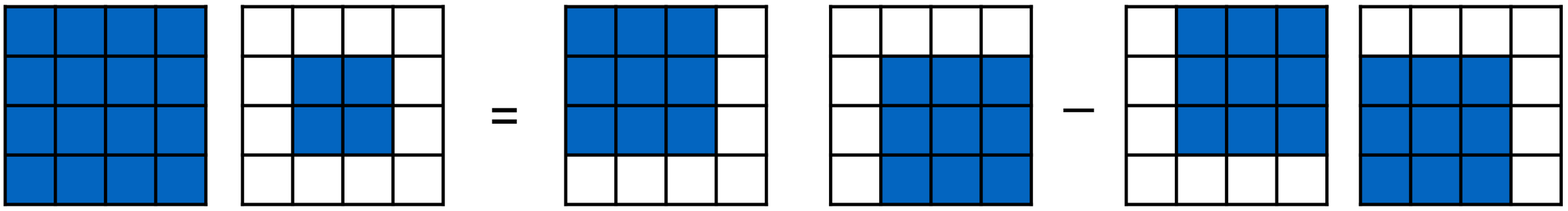


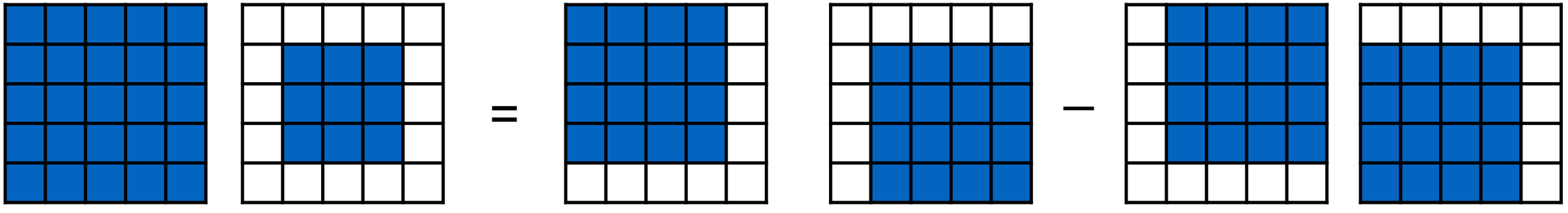
Dodgson's Rule

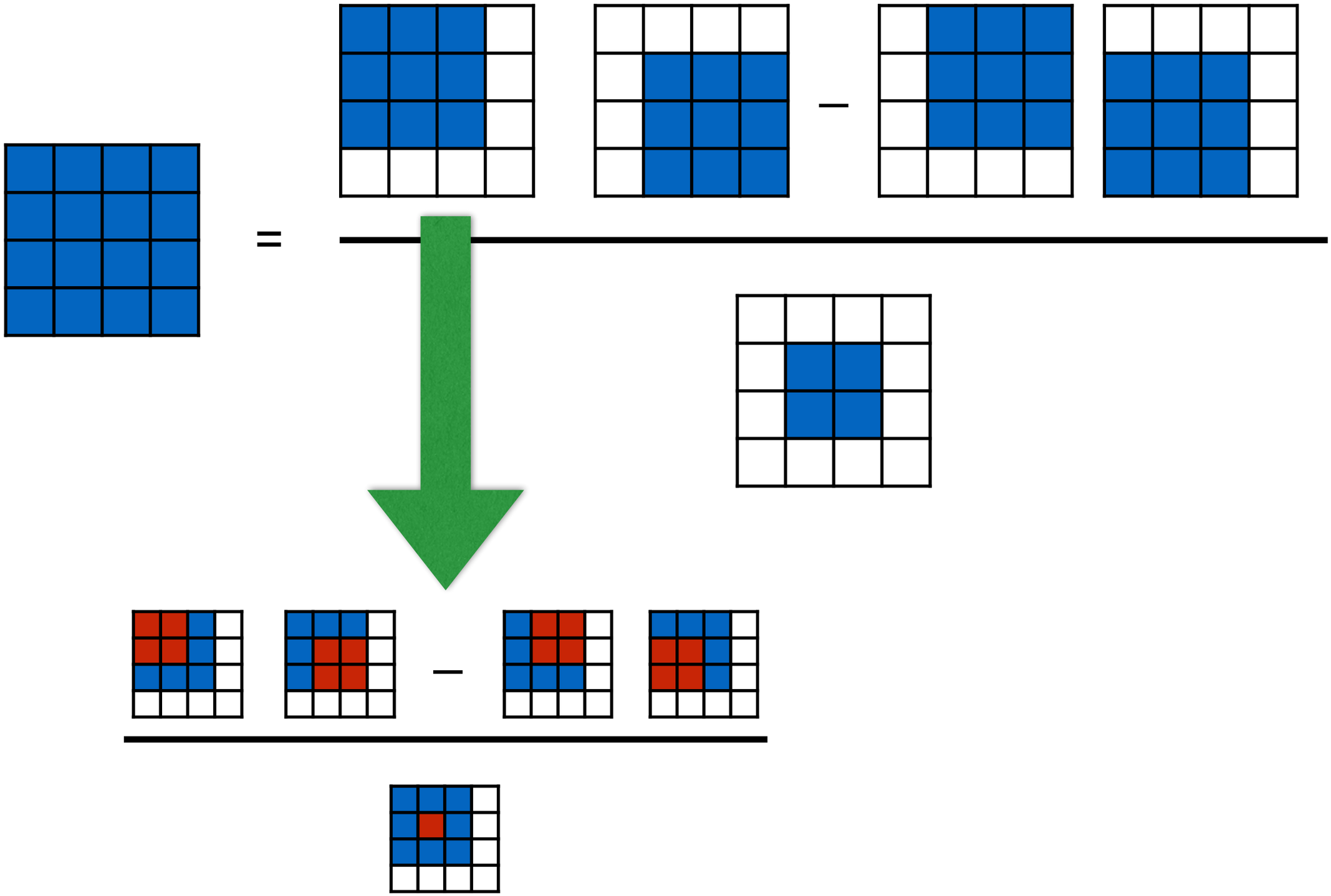
$$\det\left[(a_{i,j})_{\substack{1\le i\le n\\1\le j\le n}}\right]\cdot\det\left[(a_{i,j})_{\substack{2\le i\le n-1\\2\le j\le n-1}}\right]=$$

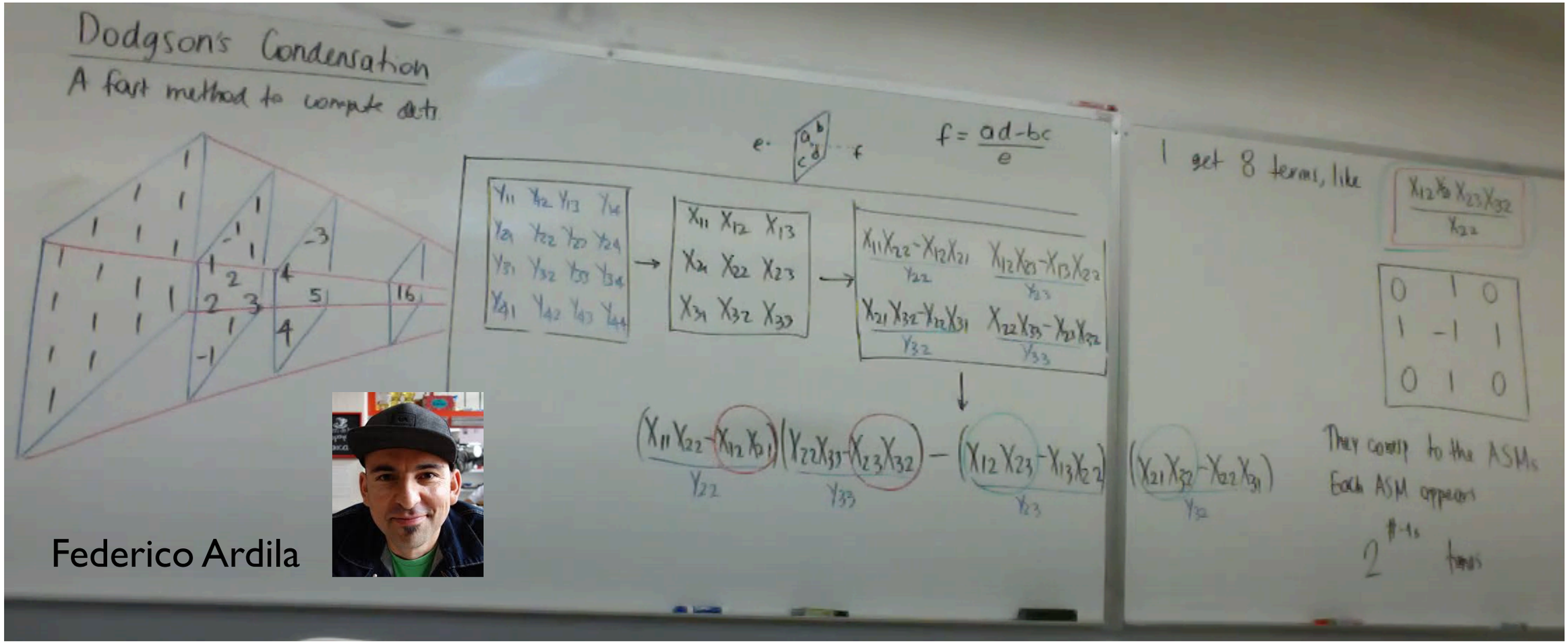
$$\det\left[(a_{i,j})_{\substack{1\le i\le n-1\\1\le j\le n-1}}\right]\cdot\det\left[(a_{i,j})_{\substack{2\le i\le n\\2\le j\le n}}\right]-\det\left[(a_{i,j})_{\substack{1\le i\le n-1\\2\le j\le n}}\right]\cdot\det\left[(a_{i,j})_{\substack{2\le i\le n\\1\le j\le n-1}}\right]$$











An alternating sign matrix is an nxn matrix of 0s, 1s, -1s such that in any row and col, the 1s and -1s alternate, starting and ending with 1.

Ex:

0	0	1	0	0
0	1	-1	0	1
1	-1	0	1	0
0	0	1	0	0
0	1	0	0	0

7 ASMs of size 3

6 are permutation matrices

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} \quad \begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$$

$$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix} \quad \begin{bmatrix}0&1&0\\1&-1&1\\0&1&0\end{bmatrix} \quad \begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$$

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix} \quad \begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$$

42 AMSs of size 4

24 are permutation matrices

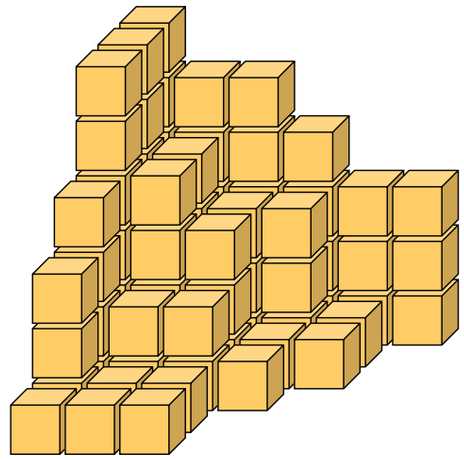
n	A_n	
1	1	
2	2	
3	7	
4	42	$= 2 \times 3 \times 7$
5	429	$= 3 \times 11 \times 13$
6	7436	$= 2^2 \times 11 \times 13^2$
7	218348	$= 2^2 \times 13^2 \times 17 \times 19$
8	10850216	$= 2^3 \times 13 \times 17^2 \times 19^2$
9	911835460	$= 2^2 \times 5 \times 17^2 \times 19^3 \times 23$

How many $n \times n$ alternating sign matrices?

The alternating sign matrix conjecture

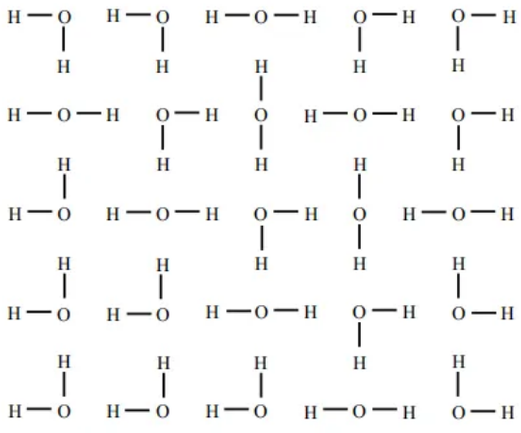
$$A_n = \prod_{j=0}^{n-1} \frac{(3j+1)!}{(n+j)!}$$

Robins and Rumsey 1986
Zeilberger 1995



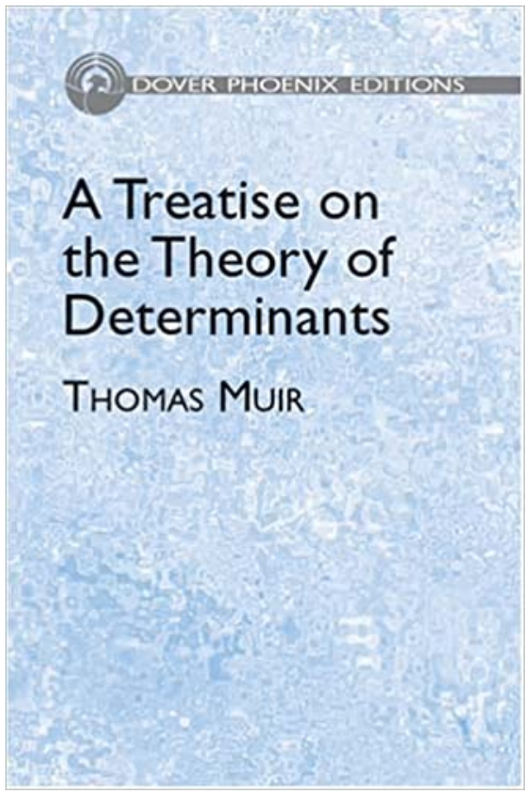
plane partition

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



square ice

Further reading



INTERNATIONAL JOURNAL OF
INFORMATION AND SYSTEMS SCIENCES
Volume 4, Number 1, Pages 1–7

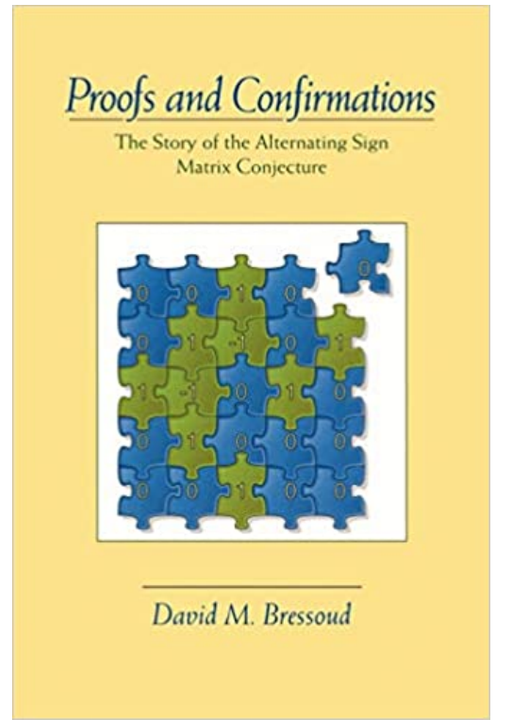
© 2008 Institute for Scientific
Computing and Information

A COMBINATORIAL PROOF OF THE DODGSON/MUIR DETERMINANTAL IDENTITY

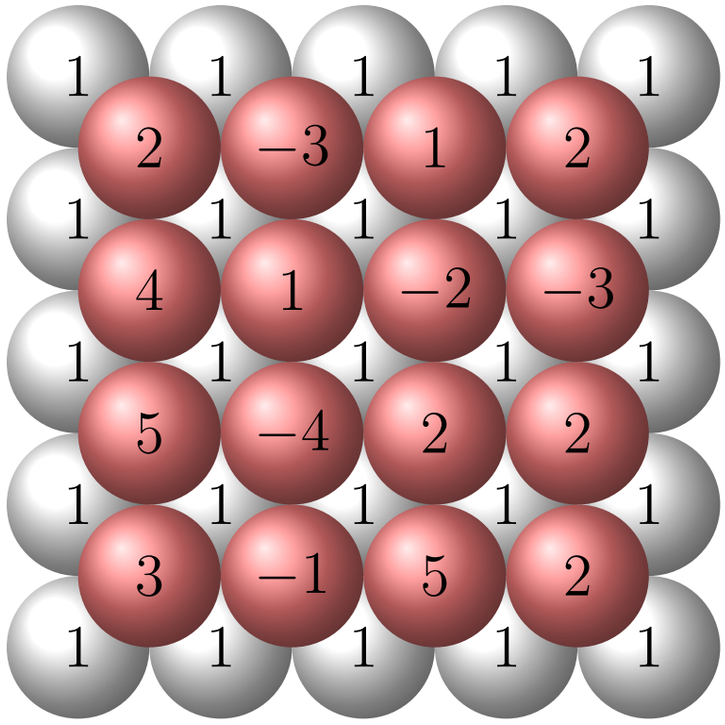
ADAM BERLINER AND RICHARD A. BRUALDI

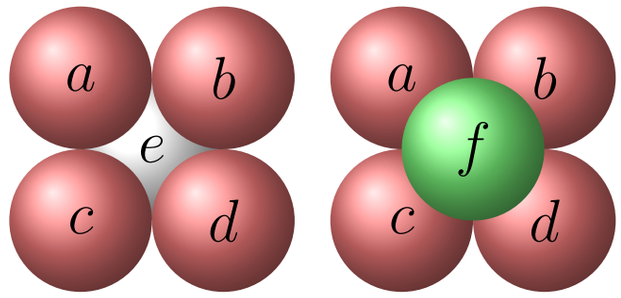
Abstract. Zeilberger has given a combinatorial proof of Dodgson's rule for calculating determinants. By use of Muir's law of extensible minors, Dodgson's rule can be generalized to an identity we call the Dodgson/Muir identity. We give a combinatorial proof of this identity.

Key Words. Determinant, determinantal identity, Dodgson's Rule, Muir's Law, combinatorial proof.

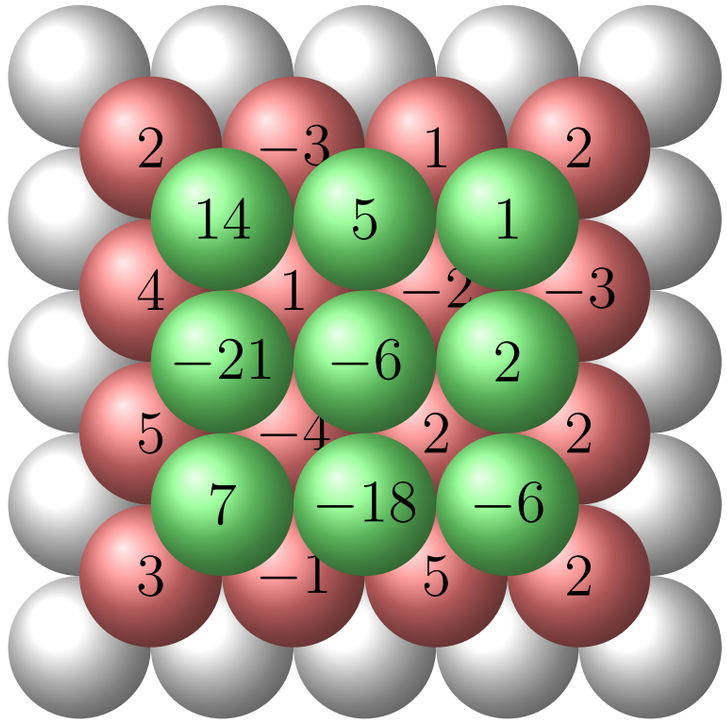


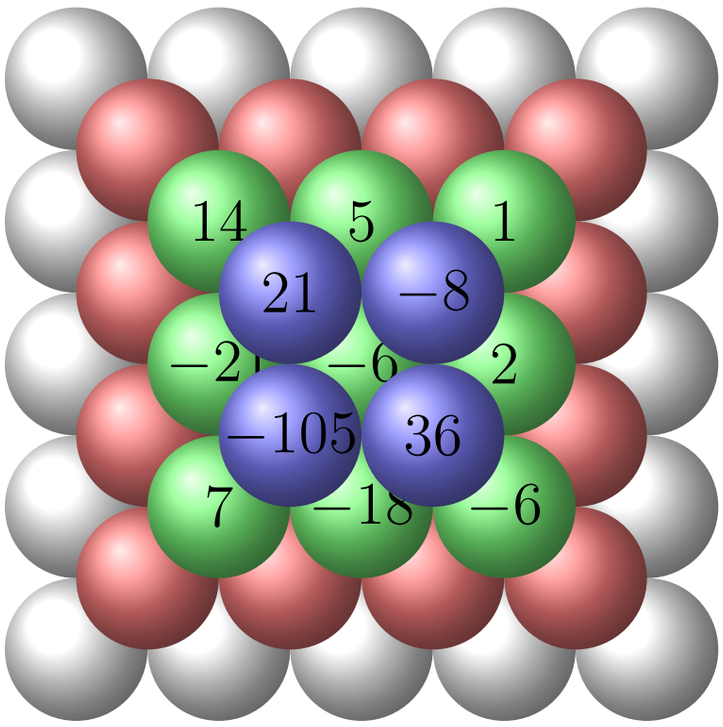
Dodgson condensation

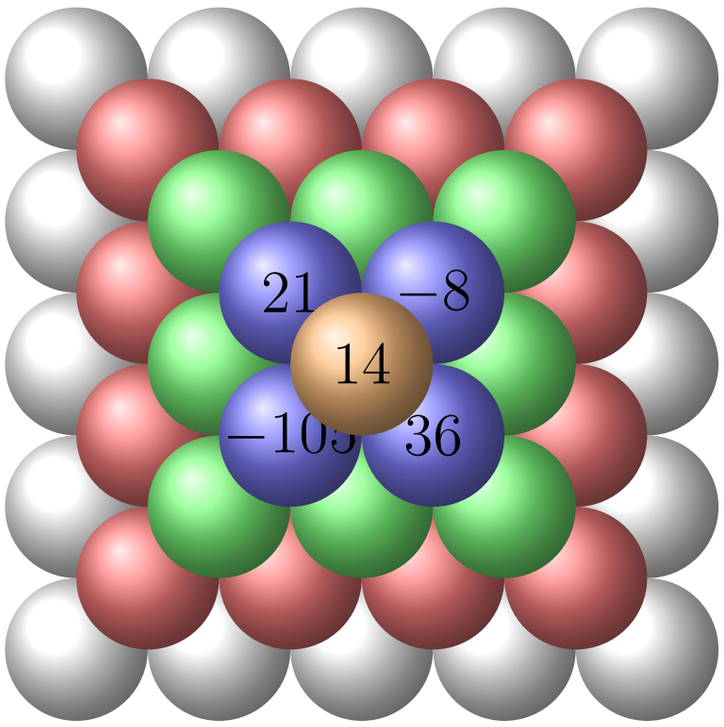




$$f = \frac{ad - bc}{e}$$







I made this using TikZ

The Routh–Hurwitz stability criterion

```
N=4;
C = sym('C', [N 1]); % C1..CN are variables representing cofactors
H = sym('M', [N N]); % Matrix HI_J of cofactors arranged as Hurwitz matrix
H=sym(zeros(N,1));
for i=1:N
    for j=1:N
        if j==2*i
            H(i,j)=1;
        elseif j<=2*i-1 && 2*i-j<=N
            H(i,j)=(-1)^j*C(2*i-j);
        end
    end
end
disp(['The Hurwitz matrix of cofactors
H
for i=1:N
    fprintf('Delta%d = ',i)
    fprintf('%s > 0',det(H(1:i,1:i)))
    if i==1
        disp([', equivalent to C1 = tr(
    else
        fprintf('\n')
    end
end
if rem(N,2)==0, str = '> 0'; else str = '< 0'; end
disp(['Given Delta' num2str(N-1) ' > 0, Delta' num2str(N) ...
    ' > 0 is equivalent to C' num2str(N) ' = det(A) ' str '.'])
return
```

$$c_2(A) = \sum_{1\le i<j\le n} \begin{vmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{vmatrix}$$

```
The Hurwitz matrix of cofactors is:

H =

[ -C1,  1,   0,  0]
[ -C3, C2, -C1,  1]
[   0, C4, -C3, C2]
[   0,  0,   0, C4]

Delta1 = -C1 > 0, equivalent to C1 = tr(A) < 0.
Delta2 = C3 - C1*C2 > 0
Delta3 = C1*C2*C3 - C3^2 - C1^2*C4 > 0
Delta4 = -C4*(C1^2*C4 + C3^2 - C1*C2*C3) > 0
Given Delta3 > 0, Delta4 > 0 is equivalent to C4 = det(A) > 0.
```

$$c_3(A) = \sum_{1\le i<j<k\le n} \begin{vmatrix} a_{ii} & a_{ij} & a_{ik} \\ a_{ji} & a_{jj} & a_{jk} \\ a_{ki} & a_{kj} & a_{kk} \end{vmatrix}$$

$$\operatorname{tr}(A) < 0\,, \quad \operatorname{tr}(A)c_2(A) < c_3(A)\,, \quad \operatorname{tr}(A)c_2(A)c_3(A) - \operatorname{tr}(A)^2 \det(A) - c_3(A)^2 > 0\,, \quad \det(A) > 0\,.$$

Turing instability in the 4×4 system

A is the 4×4 Jacobian of the F1 system in the absence of diffusion. The stability condition implies $\mathrm{tr}A = f_u + g_v + \hat{f}_{\hat{u}} + \hat{g}_{\hat{v}} < 0$ and $\det A >$ for the spatially homogenous steady state.

Let A' be the system with the the diffusion and spatial frequency term included on the diagonal,

$$A' = A - \begin{bmatrix} D_u k^2 & 0 & 0 & 0 \\ 0 & D_v k^2 & 0 & 0 \\ 0 & 0 & D_{\hat{u}} k^2 & 0 \\ 0 & 0 & 0 & D_{\hat{v}} k^2 \end{bmatrix}.$$

In this form it is difficult to see the change that occurs when diffusion is included ($\det A' - \det A$). It is clear that $\det A'$ includes $\det A$ plus a contribution from any permutation where $w_i = i$. The 24 permutations are

$$^{+}\mathbf{1234}\ ^{-}\mathbf{12}43\ ^{-}\mathbf{1}32\mathbf{4}\ ^{+}\mathbf{1}342\ ^{+}\mathbf{1}423\ ^{-}\mathbf{1}4\mathbf{3}2\ ^{-}2\mathbf{34}\ ^{+}2143\ ^{+}231\mathbf{4}\ ^{-}2341\ ^{-}2413\ ^{+}24\mathbf{3}1$$
$$^{+}312\mathbf{4}\ ^{-}3142\ ^{-}3\mathbf{2}1\mathbf{4}\ ^{+}3\mathbf{2}41\ ^{+}3412\ ^{-}3421\ ^{-}4123\ ^{+}41\mathbf{3}2\ ^{+}4\mathbf{2}13\ ^{-}4\mathbf{23}1\ ^{-}4312\ ^{+}4321\,,$$

where **Bold red** indicates a contribution of the form $D_i k^2$ (see Fig. 1). For each permutation involving one or more $D_i k^2$, one may calculate the terms that arise in $\det A'$ that are not in $\det A$. For example,

$$^{+}\mathbf{1}342 \quad \rightsquigarrow \quad -D_u g_{\hat{u}} \hat{f}_{\hat{v}} \hat{g}_v k^2$$
$$^{+}\mathbf{1}423 \quad \rightsquigarrow \quad -D_u g_{\hat{v}} \hat{f}_v \hat{g}_{\hat{u}} k^2$$
$$^{-}4\mathbf{23}1 \quad \rightsquigarrow \quad \hat{g}_{\hat{v}}(\hat{g}_u - D_u k^2)(\hat{f}_{\hat{u}} - D_{\hat{u}} k^2)\hat{g}_u \rightsquigarrow -\hat{g}_{\hat{v}}(D_u \hat{f}_{\hat{u}} + \hat{g}_u D_{\hat{u}})\hat{g}_u k^2 + \hat{g}_{\hat{v}} D_u D_{\hat{u}} \hat{g}_u k^4$$

The stable below shows the number of terms expected in $\det A' - \det A$ broken down in terms of the power of the wavenumber k:

#	permutation type	$\circ\circ\circ k^2$	$\circ\circ k^4$	$\circ k^6$	k^8
8	$(* + D_* k^2) * * *$	1			
6	$(* + D_* k^2)^2 * *$	2	1		
0	$(* + k^2)^3 *$	3	3	1	
1	$(* + k^2)^4$	4	6	4	1
		24	12	4	1

For a direct calculation of $|A'|$, recall that linearity of determinants implies identities of the form

$$\begin{vmatrix} a_1 + x_1 & b_1 & c_1 \\ a_2 + x_2 & b_2 & c_2 \\ a_3 + x_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x_1 & b_1 & c_1 \\ x_2 & b_2 & c_2 \\ x_3 & b_3 & c_3 \end{vmatrix}$$

Setting $x_2 = x_3 = 0$, we see that

$$\begin{vmatrix} a_1 + x_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x_1 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + x_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} .$$

Our calculation will use identities of this form. Writing $\kappa_i = D_i k^2$, we find

$$\begin{aligned} |A'| &= \begin{vmatrix} f_u - \kappa_u & f_v & f_{\hat{u}} & f_{\hat{v}} \\ g_u & g_v - \kappa_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} - \kappa_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_u & \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} \\ &= \begin{vmatrix} f_u & f_v & f_{\hat{u}} & f_{\hat{v}} \\ g_u & g_v - \kappa_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} - \kappa_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_u & \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} - \kappa_u \begin{vmatrix} g_v - \kappa_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_v & \hat{f}_{\hat{u}} - \kappa_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} . \end{aligned}$$

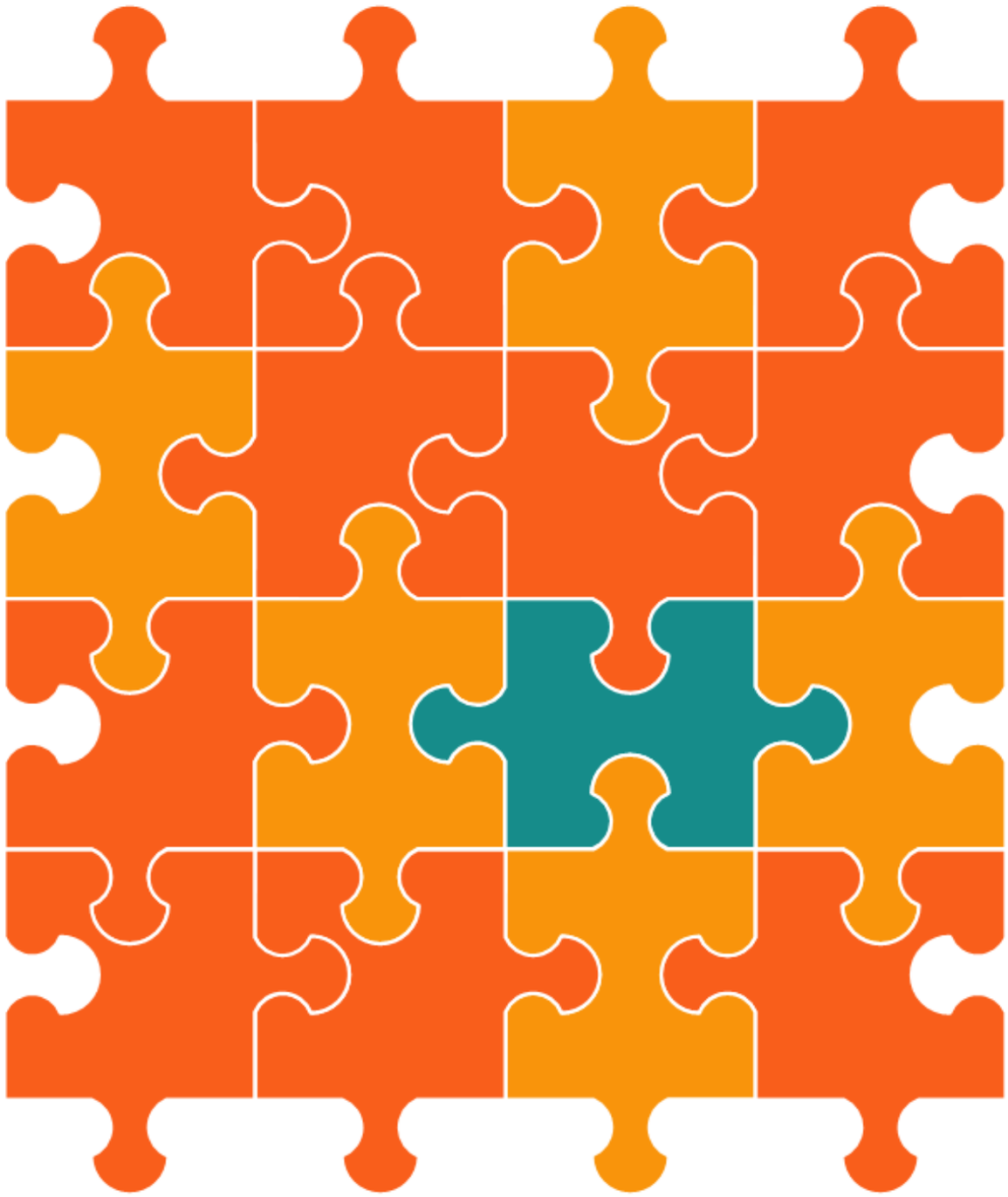
Continuing this process we get

$$
\begin{aligned}
|A'| &= |A| - \kappa_u \begin{vmatrix} g_v - \kappa_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_v & \hat{f}_{\hat{u}} - \kappa_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} - \kappa_v \begin{vmatrix} f_u & f_{\hat{u}} & f_{\hat{v}} \\ \hat{f}_u & \hat{f}_{\hat{u}} - \kappa_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_u & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} \\
&- \kappa_{\hat{u}} \begin{vmatrix} f_u & f_v & f_{\hat{v}} \\ g_u & g_v & g_{\hat{v}} \\ \hat{g}_u & \hat{g}_v & \hat{g}_{\hat{v}} - \kappa_{\hat{v}} \end{vmatrix} - \kappa_{\hat{v}} \begin{vmatrix} f_u & f_v & f_{\hat{u}} \\ g_u & g_v & g_{\hat{u}} \\ \hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} \end{vmatrix} \\
&= |A| - \kappa_u \begin{vmatrix} g_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_v & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} + \kappa_u\kappa_{\hat{v}} \begin{vmatrix} g_v & g_{\hat{u}} \\ \hat{f}_v & \hat{f}_{\hat{u}} \end{vmatrix} + \kappa_u\kappa_{\hat{u}} \begin{vmatrix} g_v & g_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{v}} \end{vmatrix} - \kappa_u\kappa_{\hat{u}}\kappa_{\hat{v}} g_v \\
&+ \kappa_u\kappa_v \begin{vmatrix} \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} - \kappa_u\kappa_v\kappa_{\hat{v}} \hat{f}_{\hat{u}} - \kappa_u\kappa_v\kappa_{\hat{u}} \hat{g}_{\hat{v}} + \kappa_u\kappa_v\kappa_{\hat{u}}\kappa_{\hat{v}} \\
&- \kappa_v \begin{vmatrix} f_u & f_{\hat{u}} & f_{\hat{v}} \\ \hat{f}_u & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_u & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} + \kappa_v\kappa_{\hat{v}} \begin{vmatrix} f_u & f_{\hat{u}} \\ \hat{f}_u & \hat{f}_{\hat{u}} \end{vmatrix} + \kappa_v\kappa_{\hat{u}} \begin{vmatrix} f_u & f_{\hat{v}} \\ \hat{g}_u & \hat{g}_{\hat{v}} \end{vmatrix} - \kappa_v\kappa_{\hat{u}}\kappa_{\hat{v}} f_u \\
&- \kappa_{\hat{u}} \begin{vmatrix} f_u & f_v & f_{\hat{v}} \\ g_u & g_v & g_{\hat{v}} \\ \hat{g}_u & \hat{g}_v & \hat{g}_{\hat{v}} \end{vmatrix} + \kappa_{\hat{u}}\kappa_{\hat{v}} \begin{vmatrix} f_u & f_v \\ g_u & g_v \end{vmatrix} \\
&- \kappa_{\hat{v}} \begin{vmatrix} f_u & f_v & f_{\hat{u}} \\ g_u & g_v & g_{\hat{u}} \\ \hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} \end{vmatrix}
\end{aligned}
$$

Using our original notation and factoring, the final answer is:

$$
\begin{aligned}
|A'| &= |A| - \left(D_u \begin{vmatrix} g_v & g_{\hat{u}} & g_{\hat{v}} \\ \hat{f}_v & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} + D_v \begin{vmatrix} f_u & f_{\hat{u}} & f_{\hat{v}} \\ \hat{f}_u & \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_u & \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} + D_{\hat{u}} \begin{vmatrix} f_u & f_v & f_{\hat{v}} \\ g_u & g_v & g_{\hat{v}} \\ \hat{g}_u & \hat{g}_v & \hat{g}_{\hat{v}} \end{vmatrix} + D_{\hat{v}} \begin{vmatrix} f_u & f_v & f_{\hat{u}} \\ g_u & g_v & g_{\hat{u}} \\ \hat{f}_u & \hat{f}_v & \hat{f}_{\hat{u}} \end{vmatrix} \right) k^2 \\
&+ \left(D_u D_{\hat{v}} \begin{vmatrix} g_v & g_{\hat{u}} \\ \hat{f}_v & \hat{f}_{\hat{u}} \end{vmatrix} + D_u D_{\hat{u}} \begin{vmatrix} g_v & g_{\hat{v}} \\ \hat{g}_v & \hat{g}_{\hat{v}} \end{vmatrix} + D_u D_v \begin{vmatrix} \hat{f}_{\hat{u}} & \hat{f}_{\hat{v}} \\ \hat{g}_{\hat{u}} & \hat{g}_{\hat{v}} \end{vmatrix} \right. \\
&+ \left. D_v D_{\hat{v}} \begin{vmatrix} f_u & f_{\hat{u}} \\ \hat{f}_u & \hat{f}_{\hat{u}} \end{vmatrix} + D_v D_{\hat{u}} \begin{vmatrix} f_u & f_{\hat{v}} \\ \hat{g}_u & \hat{g}_{\hat{v}} \end{vmatrix} + D_{\hat{u}} D_{\hat{v}} \begin{vmatrix} f_u & f_v \\ g_u & g_v \end{vmatrix} \right) k^4 \\
&- \left(D_v D_{\hat{u}} D_{\hat{v}} f_u + D_u D_{\hat{u}} D_{\hat{v}} g_v + D_u D_v D_{\hat{v}} \hat{f}_{\hat{u}} + D_u D_v D_{\hat{u}} \hat{g}_{\hat{v}} \right) k^6 + D_u D_v D_{\hat{u}} D_{\hat{v}} k^8
\end{aligned}
$$

An alternating sign matrix



INTERNATIONAL JOURNAL OF
INFORMATION AND SYSTEMS SCIENCES
Volume 4, Number 1, Pages 1–7

© 2008 Institute for Scientific
Computing and Information

A COMBINATORIAL PROOF OF THE DODGSON/MUIR DETERMINANTAL IDENTITY

ADAM BERLINER AND RICHARD A. BRUALDI

Abstract. Zeilberger has given a combinatorial proof of Dodgson's rule for calculating determinants. By use of Muir's law of extensible minors, Dodgson's rule can be generalized to an identity we call the Dodgson/Muir identity. We give a combinatorial proof of this identity.

Key Words. Determinant, determinantal identity, Dodgson's Rule, Muir's Law, combinatorial proof.

1. Introduction

Let $A = [a_{ij}]$ be a matrix of order n. For $I, J \subseteq \{1, 2, \ldots, n\}$, let $A[I, J]$ denote the submatrix of A formed by rows with index in I and columns with index in J. The complementary submatrix of $A[I, J]$ in A, that is, the submatrix formed by rows and columns with indices in the complements of I and J, respectively, is denoted by $A(I, J)$. If $I = \{i\}$ and $J = \{j\}$, then we shorten $A[I, J]$ to $A[i, j]$ and $A(I, J)$ to $A(i, j)$. Let $K = \{3, \ldots, n\}$. Dodgson's rule [2] for determinants, also known as the Desnanot-Jacobi identity (see [4, pp. 136-149] and [3, sec.3]), asserts that

$$\begin{aligned} \det A \det A[K, K] &= \det A[\{1\} \cup K, \{1\} \cup K] \det A[\{2\} \cup K, \{2\} \cup K] \\ &\quad - \det A[\{1\} \cup K, \{2\} \cup K] \det A[\{2\} \cup K, \{1\} \cup K]. \end{aligned} \tag{1}$$

Dodgson's rule can be regarded as a special case of Muir's law of extensible minors (see [1]). This law asserts that a homogeneous determinantal identity for the minors of a matrix remains valid when all the index sets are enlarged by the same disjoint index set. Consider the ordinary expansion of the determinant of a matrix $A = [a_{ij}]$ of order k given by

$$\begin{aligned} \det A &= \sum_{\sigma \in S_k} (-1)^{\iota(\sigma)} a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{k\sigma(k)} \\ &= \sum_{\sigma \in S_k} (-1)^{\iota(\sigma)} \det A[1, \sigma(1)] \det A[2, \sigma(2)] \cdots \det A[k, \sigma(k)], \end{aligned} \tag{2}$$

where $\iota(\sigma)$ is the number of inversions of the permutation σ and S_k is the set of all permutations of $\{1, 2, \ldots, k\}$. The determinants on the righthand side in (2) are determinants of matrices of order 1 and each summand is a product of k determinants. The lefthand side of (2) contains only one determinant. Making (2) into a homogeneous identity, we obtain

(3)
$$\det A \det A[\emptyset, \emptyset]^{k-1} = \sum_{\sigma \in S_k} (-1)^{\iota(\sigma)} \det A[1, \sigma(1)] \det A[2, \sigma(2)] \cdots \det A[k, \sigma(k)],$$

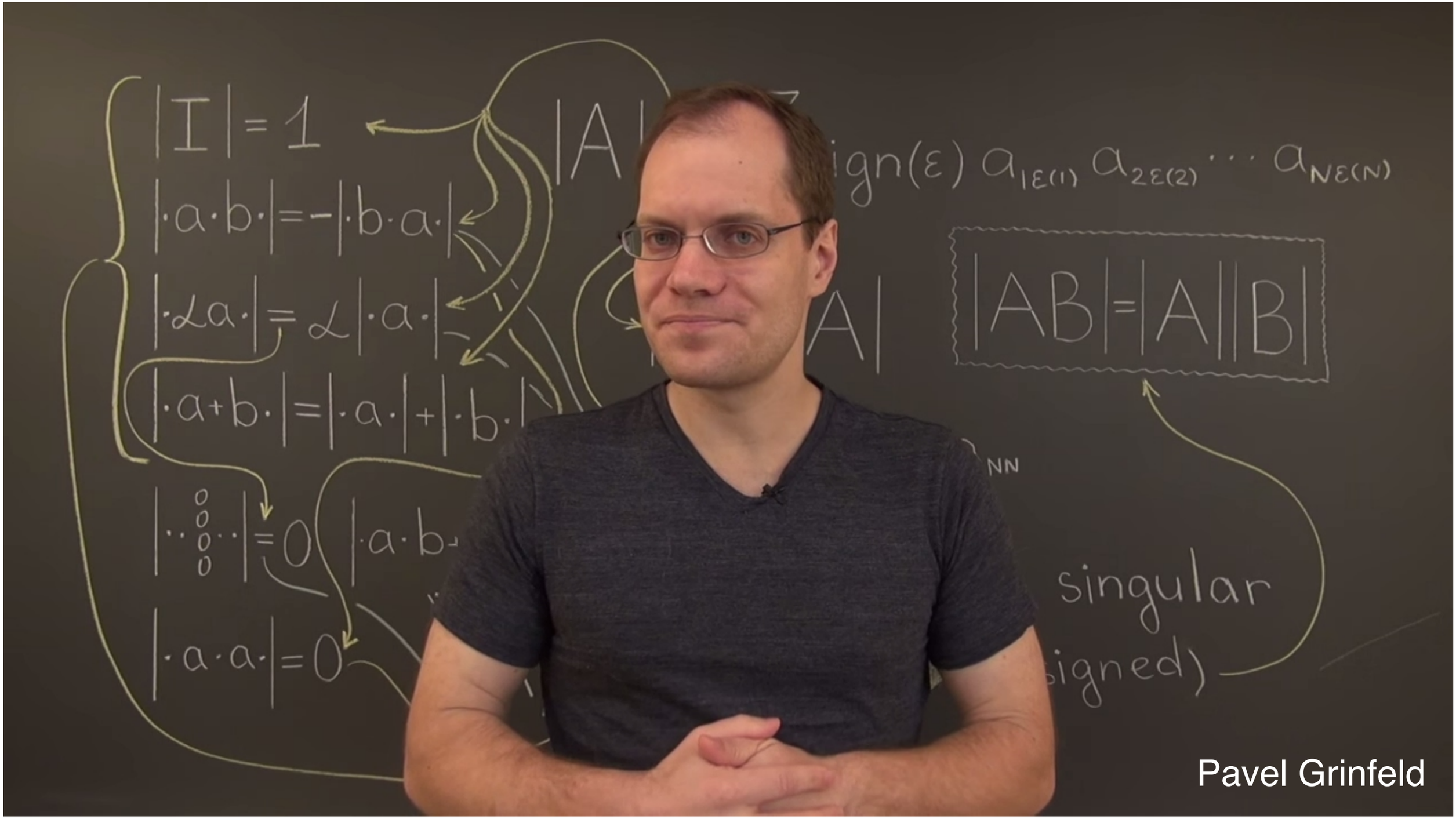
where $A[\emptyset, \emptyset]$ denotes the empty submatrix of A with determinant equal to 1. Now the lefthand side of (3) and each summand in the righthand side is a product of k determinants. Enlarging the index sets of all the determinants in (3), we obtain, with A now a matrix of order n, the *Dodgson/Muir determinantal identity*:

$$\det A \det A[\{k+1, \cdots, n\}, \{k+1, \cdots, n\}]^{k-1} =$$

$$\sum_{\sigma \in S_k} (-1)^{\iota(\sigma)} \prod_{j=1}^{k} \det A[\{j, k+1, \ldots, n\}, \{\sigma(j), k+1, \ldots, n\}]. \tag{4}$$

The special case of (4) obtained by taking $k = 2$ is Dodgson's rule (1).

If you are trying to say as much as you can about a square matrix using only one number, that number has got to be the *determinant*. —Gil Strang



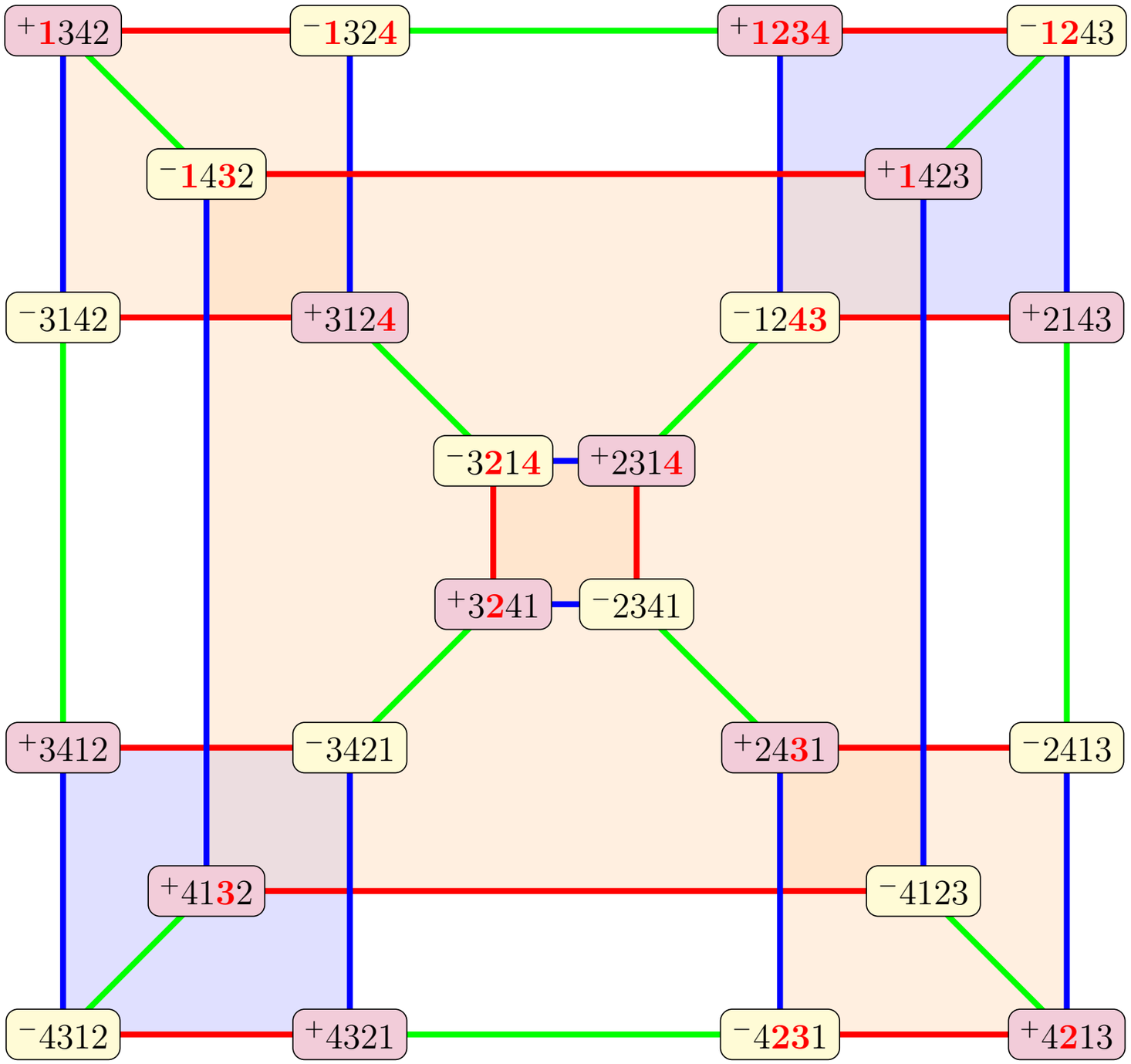
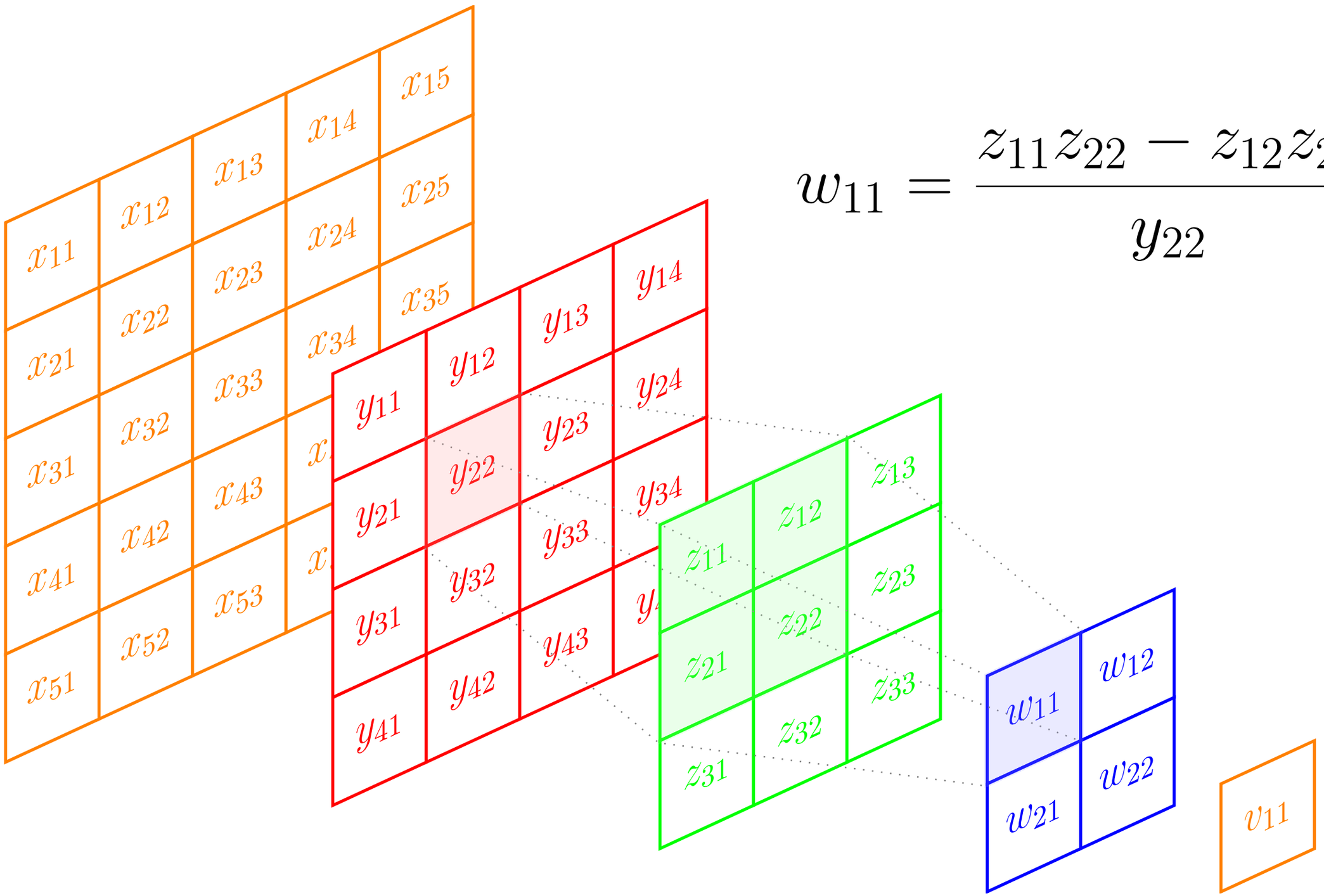


Figure 1: The order 4 permutohedron. The blue faces correspond to the terms $|A_{11}||A_{22}|$ and $|A_{12}||A_{21}|$. Even permutations are red; odd permutations are yellow. Colored edges correspond to the three generators of the symmetric group. The red bold numbers indicate $w_i = i$ (elements not moved by permutation).

Dodgson condensation

$$w_{11} = \frac{z_{11}z_{22} - z_{12}z_{21}}{y_{22}}$$



I made this using TikZ

Notes on determinants of block matrices

Best reference seems to be a 2000 paper by Silvester.

Verbatim from: https://djalil.chafai.net/blog/2012/10/14/determinant-of-block-matrices/

See also: https://arxiv.org/abs/1112.4379

Consider a block matrix of size $(n+m)\times(n+m)$ of the form

$$M=\begin{bmatrix}A & B\\ C & D\end{bmatrix}$$

where A is $n\times n$, B is $n\times m$, C is $m\times n$ and D is $m\times m$. If D is invertible then

$$\det M=\det(A-BD^{-1}C)\det D$$

If $m=n$ and C,D commute then $\det(M)=\det(AD-BC)$.